Off-Grid Inverter

SUN-6.6/8.8K-OG03LP1-EU-AM2



Home-Ready Design
Upgraded layered structure with curved panels

Built-in Wi-Fi
No extra logger needed

Multiple Power Options
3-8.8kW models to meet different energy needs

Lightweight & Easy Installation
Only 8.6 kg, enabling effortless handling and setup

Powerful Charge/Discharge Current
Up to 190A

2× PV Oversizing
Maximize energy harvest

Up to 2 MPPTs
Support high-current PV modules

200% Peak Output
Reliable high-power startup

IP65 Protection Rating
High-level dust and water resistant

Max. 97.6% Efficiency
Low charge/discharge losses

4ms On/Off-Grid Switching
Protect sensitive loads

Flexible Expansion
Max.16 inverters/ 32 batteries in parallel

Technical Data

Model	SUN-6.6K-OG03 LP1-EU-AM2	SUN-8.8K-OG03 LP1-EU-AM2
Battery Input Data		
Battery Type	Lead-acid or Lithium-ion	
Battery Voltage Range (V)	40-60	
Max. Charging/Discharging Current (A)	140	190
Charging Strategy for Li-ion Battery	Self-adaption to BMS	
Number of Battery Input	1	
PV String Input Data		
Max. PV access power (W)	13200	17600
Max. PV Input Power (W)	10560	14080
Max. PV Input Voltage (V)	500	
Start-up Voltage (V)	125	
MPPT Voltage Range (V)	150-425	
Rated PV Input Voltage (V)	370	
Max. Operating PV Input Current (A)	18+18	36+18
Max. Input Short-Circuit Current (A)	27+27	54+27
No. of MPP Trackers/No. of Strings MPP Tracker	2/1+1	2/2+1
AC Output Data		
Rated AC Output Power (VA/W)	6600	8800
Max. AC Output Power (VA/W)	6600	8800
Max. AC Output Current (A)	28.7	38.3
Rated AC Output Current (A)	28.7	38.3
Peak Power (W)	2 times of rated power, 10s	
Rated output voltage	230 L+N+PE	
Rated Output Frequency	50Hz/60Hz	
Output Voltage Waveform	Pure Sine Wave	
Total Current Harmonic Distortion THDi	<3%	
AC Input Data(Grid and Generator)		
Max. Input Power to Battery (W)	6600	8800
Rated Input Voltage	230	
Rated Input Frequency	50Hz/60Hz	
Grid Input Current (A)	35	50
Generator Input Current (A)	35	50
Efficiency		
Max. Efficiency	97.6%	
Euro Efficiency	96.5%	
MPPT Efficiency	>99%	
Equipment Protection		
Integrated	DC Reverse Polarity Protection, AC Output Overcurrent Protection, Thermal Protection, AC Output Overvoltage Protection, AC Output Short Circuit Protection, DC Component Monitoring, Insulation Impedance Detection	
Surge Protection Level	TYPE II(DC), TYPE II(AC)	
Interface		
LCD/LED Display	LCD	
Communication Interface	Wi-Fi&BT,RS485,CAN	
General Data		
Operating Temperature Range (°C)	-40 to +60°C, >45°C Derating	
Permissible Ambient Humidity	0-100%	
Permissible Altitude	3000m	
Noise (dB)	<46	
Ingress Protection(IP) Rating	IP 65	
Inverter Topology	Non-Isolated (solar), Isolated (battery)	
Over Voltage Category	OVC II(DC), OVC III(AC)	
Cabinet Size (WxHxD mm)	323×424×176 (Excluding Connectors and Brackets)	
Weight (kg)	8.6	
Type of Cooling	Intelligent Air Cooling	
Warranty	5 Years	
Safety / EMC Standard	IEC62109-1/-2, EN61000-6-1,EN61000-6-2,EN61000-6-3,EN61000-6-4	

Single Phase Hybrid Inverter

SUN-10/12K-SG02LP1-EU-AM3-P



Intuitive

- Colorful touch LCD, IP65 protection degree

Intelligent

- 6 time periods for battery charging/discharging

Flexible

- AC couple to retrofit existing solar system
- Max. 16 pcs parallel for on-grid and off-grid operation; support multiple batteries parallel

Reliable

- Max. charging/discharging current of 250A
- Support storing energy from diesel generator

Technical Data

Model	SUN-10K-SG02 LP1-EU-AM3-P	SUN-12K-SG02 LP1-EU-AM3-P
Battery Input Data		
Battery Type	Lead-acid or Lithium-ion	
Battery Voltage Range (V)	40-60	
Max. Charging/Discharging Current (A)	220	250
Charging Strategy for Li-ion Battery	Self-adaption to BMS	
Number of Battery Input	1	
PV String Input Data		
Max. PV access power (W)	20000	24000
Max. PV Input Power (W)	16000	19200
Max. PV Input Voltage (V)	500	
Start-up Voltage (V)	125	
MPPT Voltage Range (V)	150-425	
Rated PV Input Voltage (V)	370	
Max. Operating PV Input Current (A)	32+32+32	
Max. Input Short-Circuit Current (A)	60+60+60	
No. of MPP Trackers/ No. of Strings MPP Tracker	3/2+2+2	
AC Input/Output Data		
Rated AC Input/Output Active Power (W)	10000	12000
Max. AC Input/Output Apparent Power (VA)	11000	13200
Rated AC Input/Output Current (A)	45.5/43.5	54.6/52.2
Max. AC Input/Output Current (A)	50/47.9	60/57.4
Max. Continuous AC Passthrough (grid to load) (A)	60	
Peak Power (off-grid) (W)	2 times of rated power, 10s	
Power Factor Adjustment Range	0.8 leading to 0.8 lagging	
Rated input/output voltage	220/230V 0.85Un-1.1Un L+N+PE	
Rated Input/Output Grid Frequency/Range(Hz)	50/45-55, 60/55-65	
Total Current Harmonic Distortion THDi	<3% (of nominal power)	
DC Injection Current	<0.5% In	
Efficiency		
Max. Efficiency	97.6%	
Euro Efficiency	96.5%	
MPPT Efficiency	>99%	
Equipment Protection		
Integrated	DC Reverse Polarity Protection, AC Output Overcurrent Protection, Thermal Protection, AC Output Overvoltage Protection, AC Output Short Circuit Protection, DC Component Monitoring, Arc Fault Circuit Interrupter (AFCI) (optional), Anti-islanding Protection, DC Switch, Insulation Impedance Detection, Residual Current Detection	
Surge Protection Level	TYPE II(DC), TYPE II(AC)	
Interface		
LCD/LED Display	LCD	
Communication Interface	WIFI, RS485, CAN	
General Data		
Operating Temperature Range (°C)	-40 to +60°C, >45°C Derating	
Permissible Ambient Humidity	0-100%	
Permissible Altitude	2000m	
Noise (dB)	<45	
Ingress Protection(IP) Rating	IP 65	
Inverter Topology	Non-Isolated (solar), Isolated (battery)	
Over Voltage Category	OVC II(DC), OVC III(AC)	
Cabinet Size (WxHxD mm)	420×670×233 (Excluding Connectors and Brackets)	
Weight (kg)	35.6	
Type of Cooling	Intelligent Air Cooling	
Warranty	5 Years / Up to 10 Years* *Warranty period varies by installation country. Refer to the Warranty Policy for details.	
Grid Regulation	IEC 61727, IEC 62116, CEI 0-21, EN 50549, NRS 097, RD 140, UNE 217002, OVE-Richtlinie R25, G98, G99, VDE-AR-N 4105	
Safety / EMC Standard	IEC/EN 61000-6-1/2/3/4, IEC/EN 62109-1, IEC/EN 62109-2	

Single Phase Hybrid Inverter

SUN-12K-SG06LP1-EU-BM3



Intuitive

- Colorful touch LCD, IP65 protection degree

Intelligent

- 6 time periods for battery charging/discharging

Flexible

- AC couple to retrofit existing solar system
- Max. 16 pcs parallel for on-grid and off-grid operation; support multiple batteries parallel

Reliable

- Max. charging/discharging current of 250A
- Support storing energy from diesel generator

Technical Data

Model	SUN-12K-SG06 LP1-EU-BM3
Battery Input Data	
Battery Type	Lead-acid or Lithium-ion
Battery Voltage Range (V)	40-60
Max. Charging/Discharging Current (A)	250
Charging Strategy for Li-ion Battery	Self-adaption to BMS
Number of Battery Input	1
PV String Input Data	
Max. PV access power (W)	24000
Max. PV Input Power (W)	19200
Max. PV Input Voltage (V)	500
Start-up Voltage (V)	125
MPPT Voltage Range (V)	150-425
Rated PV Input Voltage (V)	370
Max. Operating PV Input Current (A)	36+36+18
Max. Input Short-Circuit Current (A)	54+54+27
No. of MPP Trackers/ No. of Strings MPP Tracker	3/2+2+1
AC Input/Output Data	
Rated AC Input/Output Active Power (W)	12000
Max. AC Input/Output Apparent Power (VA)	13200
Rated AC Input/Output Current (A)	54.6/52.2
Max. AC Input/Output Current (A)	60/57.4
Max. Continuous AC Passthrough (grid to load) (A)	70
Peak Power (off-grid) (W)	2 times of rated power, 10s
Power Factor Adjustment Range	0.8 leading to 0.8 lagging
Rated input/output voltage	220/230V 0.85Un-1.1Un L+N+PE
Rated Input/Output Grid Frequency/Range(Hz)	50/45-55, 60/55-65
Total Current Harmonic Distortion THDi	<3% (of nominal power)
DC Injection Current	<0.5% In
Efficiency	
Max. Efficiency	97.6%
Euro Efficiency	96.5%
MPPT Efficiency	>99%
Equipment Protection	
Integrated	DC Reverse Polarity Protection, AC Output Overcurrent Protection, Thermal Protection, AC Output Overvoltage Protection, AC Output Short Circuit Protection, DC Component Monitoring, Arc Fault Circuit Interrupter (AFCI) (optional), Anti-islanding Protection, DC Switch, Insulation Impedance Detection, Residual Current Detection
Surge Protection Level	TYPE II(DC), TYPE II(AC)
Interface	
LCD/LED Display	LCD
Communication Interface	WIFI, RS485, CAN
General Data	
Operating Temperature Range (°C)	-40 to +60°C, >45°C Derating
Permissible Ambient Humidity	0-100%
Permissible Altitude	3000m
Noise (dB)	<45
Ingress Protection(IP) Rating	IP 65
Inverter Topology	Non-Isolated (solar), Isolated (battery)
Over Voltage Category	OVC II(DC), OVC III(AC)
Cabinet Size (WxHxD mm)	340×580×227 (Excluding Connectors and Brackets)
Weight (kg)	24.5
Type of Cooling	Intelligent Air Cooling
Warranty	5 Years / Up to 10 Years* *Warranty period varies by installation country. Refer to the Warranty Policy for details.
Grid Regulation	IEC 61727, IEC 62116, CEI 0-21, EN 50549, NRS 097, RD 140, UNE 217002, OVE-Richtlinie R25, G99, VDE-AR-N 4105
Safety / EMC Standard	IEC/EN 61000-6-1/2/3/4, IEC/EN 62109-1, IEC/EN 62109-2

Three Phase Hybrid Inverter

SUN-12K-SG05LP3-EU-SM2



Safe

- 48V low voltage battery, transformer isolation design

Intelligent

- 6 time periods for battery charging/discharging

Flexible

- AC couple to retrofit existing solar system
- Max. 10 pcs parallel for on-grid and off-grid operation; support multiple batteries parallel

Reliable

- Max. charging/discharging current of 240A
- Support storing energy from diesel generator
- 100% unbalanced output, max. output up to 50% rated power for each phase

Technical Data

Model	SUN-12K-SG05 LP3-EU-SM2
Battery Input Data	
Battery Type	Lead-acid or Lithium-ion
Battery Voltage Range (V)	40-60
Max. Charging/Discharging Current (A)	240
Charging Strategy for Li-ion Battery	Self-adaption to BMS
Number of Battery Input	1
PV String Input Data	
Max. PV access power (W)	24000
Max. PV Input Power (W)	19200
Max. PV Input Voltage (V)	800
Start-up Voltage (V)	160
MPPT Voltage Range (V)	200-650
Rated PV Input Voltage (V)	550
Max. Operating PV Input Current (A)	26+26
Max. Input Short-Circuit Current (A)	39+39
No. of MPP Trackers/ No. of Strings MPP Tracker	2/2+2
AC Input/Output Data	
Rated AC Input/Output Active Power (W)	12000
Max. AC Input/Output Apparent Power (VA)	13200
Rated AC Input/Output Current (A)	18.2/17.4
Max. AC Input/Output Current (A)	20/19.2
Max. Continuous AC Passthrough (grid to load) (A)	45
Peak Power (off-grid) (W)	2 times of rated power, 10s
Power Factor Adjustment Range	0.8 leading to 0.8 lagging
Rated input/output voltage	220/380V,230/400V 0.85Un-1.1Un 3L+N+PE
Rated Input/Output Grid Frequency/Range(Hz)	50/45-55, 60/55-65
Total Current Harmonic Distortion THDi	<3% (of nominal power)
DC Injection Current	<0.5% In
Efficiency	
Max. Efficiency	97.6%
Euro Efficiency	97.0%
MPPT Efficiency	>99%
Equipment Protection	
Integrated	DC Reverse Polarity Protection, AC Output Overcurrent Protection, Thermal Protection, AC Output Overvoltage Protection, AC Output Short Circuit Protection, DC Component Monitoring, Arc Fault Circuit Interrupter (AFCI)(optional), Anti-islanding Protection, DC Switch, Insulation Impedance Detection, Residual Current Detection
Surge Protection Level	TYPE II(DC), TYPE II(AC)
Interface	
LCD/LED Display	LCD
Communication Interface	WIFI,RS485,CAN
General Data	
Operating Temperature Range (°C)	-40 to +60°C, >45°C Derating
Permissible Ambient Humidity	0-100%
Permissible Altitude	3000m
Noise (dB)	≤55
Ingress Protection(IP) Rating	IP 65
Inverter Topology	Non-Isolated (solar), Isolated (battery)
Over Voltage Category	OVC II(DC), OVC III(AC)
Cabinet Size (WxHxD mm)	386×660×250 (Excluding Connectors and Brackets)
Weight (kg)	35.2
Type of Cooling	Intelligent Air Cooling
Warranty	5 Years / Up to 10 Years* *Warranty period varies by installation country. Refer to the Warranty Policy for details.
Grid Regulation	IEC 61727, IEC 62116, CEI 0-21, EN 50549, NRS 097, RD 140, UNE 217002, OVE-Richtlinie R25, G99, G98, VDE-AR-N 4105
Safety / EMC Standard	IEC/EN 61000-6-1/2/3/4, IEC/EN 62109-1, IEC/EN 62109-2

Single Phase Hybrid Inverter

SUN-16K-SG01LP1-EU-AM3-P



Intuitive

- Colorful touch LCD, IP65 protection degree

Intelligent

- 6 time periods for battery charging/discharging

Flexible

- AC couple to retrofit existing solar system
- Max. 16 pcs parallel for on-grid and off-grid operation; support multiple batteries parallel

Reliable

- Max. charging/discharging current of 380A
- Support storing energy from diesel generator

Technical Data

Model	SUN-18K-SG01 LP1-EU-AM3-P
Battery Input Data	
Battery Type	Lead-acid or Lithium-ion
Battery Voltage Range (V)	40-60
Max. Charging/Discharging Current (A)	380
Charging Strategy for Li-ion Battery	Self-adaption to BMS
Number of Battery Input	2
PV String Input Data	
Max. PV access power (W)	36000
Max. PV Input Power (W)	28800
Max. PV Input Voltage (V)	500
Start-up Voltage (V)	125
MPPT Voltage Range (V)	150-425
Rated PV Input Voltage (V)	370
Max. Operating PV Input Current (A)	36+36+36
Max. Input Short-Circuit Current (A)	54+54+54
No. of MPP Trackers/ No. of Strings MPP Tracker	3/2+2+2
AC Input/Output Data	
Rated AC Input/Output Active Power (W)	18000
Max. AC Input/Output Apparent Power (VA)	19800
Rated AC Input/Output Current (A)	81.9/78.3
Max. AC Input/Output Current (A)	90/86.1
Max. Continuous AC Passthrough (grid to load) (A)	200
Peak Power (off-grid) (W)	2 times of rated power, 10s
Power Factor Adjustment Range	0.8 leading to 0.8 lagging
Rated input/output voltage	220/230V 0.85Un-1.1Un L+N+PE
Rated Input/Output Grid Frequency/Range(Hz)	50/45-55, 60/55-65
Total Current Harmonic Distortion THDi	<3% (of nominal power)
DC Injection Current	<0.5% In
Efficiency	
Max. Efficiency	97.6%
Euro Efficiency	96.5%
MPPT Efficiency	>99%
Equipment Protection	
Integrated	DC Reverse Polarity Protection, AC Output Overcurrent Protection, Thermal Protection, AC Output Overvoltage Protection, AC Output Short Circuit Protection, DC Component Monitoring, Arc Fault Circuit Interrupter (AFCI) (optional), Anti-islanding Protection, DC Switch, Insulation Impedance Detection, Residual Current Detection
Surge Protection Level	TYPE II(DC), TYPE II(AC)
Interface	
LCD/LED Display	LCD
Communication Interface	WIFI, RS485, CAN
General Data	
Operating Temperature Range (°C)	-40 to +60°C, >45°C Derating
Permissible Ambient Humidity	0-100%
Permissible Altitude	3000m
Noise (dB)	<55
Ingress Protection(IP) Rating	IP 65
Inverter Topology	Non-Isolated (solar), Isolated (battery)
Over Voltage Category	OVC II(DC), OVC III(AC)
Cabinet Size (WxHxD mm)	464×863×282
Weight (kg)	59.8
Type of Cooling	Intelligent Air Cooling
Warranty	5 Years / Up to 10 Years* *Warranty period varies by installation country. Refer to the Warranty Policy for details.
Grid Regulation	IEC 61727, IEC 62116, AS 4777.2, NRS 097
Safety / EMC Standard	IEC/EN 61000-6-1/2/3/4, IEC/EN 62109-1, IEC/EN 62109-2

Three Phase Hybrid Inverter

SUN-16/20K-SG05LP3-EU-SM2



Safe

- 48V low voltage battery, transformer isolation design

Intelligent

- 6 time periods for battery charging/discharging

Flexible

- AC couple to retrofit existing solar system
- Max. 10 pcs parallel for on-grid and off-grid operation; support multiple batteries parallel

Reliable

- Max. charging/discharging current of 350A
- Support storing energy from diesel generator
- 100% unbalanced output, max. output up to 50% rated power for each phase

Technical Data

Model	SUN-16K-SG05 LP3-EU-SM2	SUN-20K-SG05 LP3-EU-SM2
Battery Input Data		
Battery Type	Lead-acid or Lithium-ion	
Battery Voltage Range (V)	40-60	
Max. Charging/Discharging Current (A)	300	350
Charging Strategy for Li-ion Battery	Self-adaption to BMS	
Number of Battery Input	2	
PV String Input Data		
Max. PV access power (W)	32000	40000
Max. PV Input Power (W)	25600	32000
Max. PV Input Voltage (V)	800	
Start-up Voltage (V)	160	
MPPT Voltage Range (V)	160-650	
Rated PV Input Voltage (V)	550	
Max. Operating PV Input Current (A)	36+36	
Max. Input Short-Circuit Current (A)	54+54	
No. of MPP Trackers/ No. of Strings MPP Tracker	2/2+2	
AC Input/Output Data		
Rated AC Input/Output Active Power (W)	16000	20000
Max. AC Input/Output Apparent Power (VA)	17600	22000
Rated AC Input/Output Current (A)	24.3/23.2	30.4/29
Max. AC Input/Output Current (A)	26.7/25.6	33.4/31.9
Max. Continuous AC Passthrough (grid to load) (A)	70	
Peak Power (off-grid) (W)	2 times of rated power, 10s	
Power Factor Adjustment Range	0.8 leading to 0.8 lagging	
Rated input/output voltage	220/380V,230/400V 0.85Un-1.1Un 3L+N+PE	
Rated Input/Output Grid Frequency/Range(Hz)	50/45-55, 60/55-65	
Total Current Harmonic Distortion THDi	<3% (of nominal power)	
DC Injection Current	<0.5% In	
Efficiency		
Max. Efficiency	97.6%	
Euro Efficiency	97.0%	
MPPT Efficiency	>99%	
Equipment Protection		
Integrated	DC Reverse Polarity Protection, AC Output Overcurrent Protection, Thermal Protection, AC Output Overvoltage Protection, AC Output Short Circuit Protection, DC Component Monitoring, Arc Fault Circuit Interrupter (AFCI)(optional), Anti-islanding Protection, DC Switch, Insulation Impedance Detection, Residual Current Detection	
Surge Protection Level	TYPE II(DC), TYPE II(AC)	
Interface		
LCD/LED Display	LCD	
Communication Interface	WIFI, RS485, CAN	
General Data		
Operating Temperature Range (°C)	-40 to +60°C, >45°C Derating	
Permissible Ambient Humidity	0-100%	
Permissible Altitude	3000m	
Noise (dB)	<60	
Ingress Protection(IP) Rating	IP 65	
Inverter Topology	Non-Isolated (solar), Isolated (battery)	
Over Voltage Category	OVC II(DC), OVC III(AC)	
Cabinet Size (WxHxD mm)	456×750×268.5 (Excluding Connectors and Brackets)	
Weight (kg)	51.9	
Type of Cooling	Intelligent Air Cooling	
Warranty	5 Years / Up to 10 Years* *Warranty period varies by installation country. Refer to the Warranty Policy for details.	
Grid Regulation	IEC 61727, IEC 62116, CEI 0-21, EN 50549, NRS 097, RD 140, UNE 217002, OVE-Richtlinie R25, G99, VDE-AR-N 4105	
Safety / EMC Standard	IEC/EN 61000-6-1/2/3/4, IEC/EN 62109-1, IEC/EN 62109-2	

Three Phase Hybrid Inverter

SUN-20/25K-SG01HP3-EU-AM2



Efficient

- High voltage battery, higher efficiency

Intelligent

- 6 time periods for battery charging/discharging

Flexible

- AC couple to retrofit existing solar system
- Max. 10 pcs parallel for on-grid and off-grid operation; support multiple batteries parallel

Reliable

- Max. charging/discharging current of 50A
- Support storing energy from diesel generator
- 100% unbalanced output

Technical Data

Model	SUN-20K-SG01 HP3-EU-AM2	SUN-25K-SG01 HP3-EU-AM2
Battery Input Data		
Battery Type	Lithium-ion	
Battery Voltage Range (V)	160-700	
Max. Charging/Discharging Current (A)	37	50
Charging Strategy for Li-ion Battery	Self-adaption to BMS	
Number of Battery Input	1	
PV String Input Data		
Max. PV access power (W)	40000	50000
Max. PV Input Power (W)	32000	40000
Max. PV Input Voltage (V)	1000	
Start-up Voltage (V)	180	
MPPT Voltage Range (V)	150-850	
Rated PV Input Voltage (V)	600	700
Max. Operating PV Input Current (A)	26+26	
Max. Input Short-Circuit Current (A)	39+39	
No. of MPP Trackers/ No. of Strings MPP Tracker	2/2+2	
AC Input/Output Data		
Rated AC Input/Output Active Power (W)	20000	25000
Max. AC Input/Output Apparent Power (VA)	22000	27500
Rated AC Input/Output Current (A)	30.4/29.0	37.9/36.3
Max. AC Input/Output Current (A)	33.4/31.9	41.7/39.9
Max. Continuous AC Passthrough (grid to load) (A)	80	
Peak Power (off-grid) (W)	1.5 times of rated power, 10s	
Power Factor Adjustment Range	0.8 leading to 0.8 lagging	
Rated input/output voltage	220/380V,230/400V 0.85Un-1.1Un 3L+N+PE	
Rated Input/Output Grid Frequency/Range(Hz)	50/45-55, 60/55-65	
Total Current Harmonic Distortion THDi	<3% (of nominal power)	
DC Injection Current	<0.5% In	
Efficiency		
Max. Efficiency	97.6%	
Euro Efficiency	97.0%	
MPPT Efficiency	>99%	
Equipment Protection		
Integrated	DC Reverse Polarity Protection, AC Output Overcurrent Protection, Thermal Protection, AC Output Overvoltage Protection, AC Output Short Circuit Protection, DC Component Monitoring, Arc Fault Circuit Interrupter (AFCI) (optional), Anti-islanding Protection, DC Switch, Insulation Impedance Detection, Residual Current Detection	
Surge Protection Level	TYPE II(DC), TYPE II(AC)	
Interface		
LCD/LED Display	LCD	
Communication Interface	WiFi, RS485, CAN	
Monitor Mode	GPRS/WIFI/Bluetooth/4G	
General Data		
Operating Temperature Range (°C)	-40 to +60°C, >45°C Derating	
Permissible Ambient Humidity	0-100%	
Permissible Altitude	2000m	
Noise (dB)	≤55	
Ingress Protection(IP) Rating	IP 65	
Inverter Topology	Non-Isolated	
Over Voltage Category	OVC II(DC), OVC III(AC)	
Cabinet Size (WxHxD mm)	408×638×237 (Excluding Connectors and Brackets)	
Weight (kg)	30.5	
Type of Cooling	Intelligent Air Cooling	
Warranty	5 Years / Up to 10 Years* *Warranty period varies by installation country. Refer to the Warranty Policy for details.	
Grid Regulation	IEC 61727, IEC 62116, CEI 0-21, EN 50549, NRS 097, RD 140, UNE 217002, OVE-Richtlinie R25, G99, VDE-AR-N 4105	
Safety / EMC Standard	IEC/EN 61000-6-1/2/3/4, IEC/EN 62109-1, IEC/EN 62109-2	

Three Phase Hybrid Inverter

SUN-30K-SG02HP3-EU-AM3



Efficient

- High voltage battery, higher efficiency

Intelligent

- 6 time periods for battery charging/discharging

Flexible

- AC couple to retrofit existing solar system
- Max. 10 pcs parallel for on-grid and off-grid operation; support multiple batteries parallel

Reliable

- Max. charging/discharging current of 75A
- Support storing energy from diesel generator
- 100% unbalanced output

Technical Data

Model	SUN-30K-SG02HP3-EU-AM3
Battery Input Data	
Battery Type	Lithium-ion
Battery Voltage Range (V)	160-700
Max. Charging/Discharging Current (A)	75
Charging Strategy for Li-ion Battery	Self-adaption to BMS
Number of Battery Input	1
PV String Input Data	
Max. PV access power (W)	60000
Max. PV Input Power (W)	48000
Max. PV Input Voltage (V)	1000
Start-up Voltage (V)	180
MPPT Voltage Range (V)	150-850
Rated PV Input Voltage (V)	600
Max. Operating PV Input Current (A)	36+36+36
Max. Input Short-Circuit Current (A)	54+54+54
No. of MPP Trackers/ No. of Strings MPP Tracker	3/2+2+2
AC Input/Output Data	
Rated AC Input/Output Active Power (W)	30000
Max. AC Input/Output Apparent Power (VA)	33000
Rated AC Input/Output Current (A)	45.5/43.5
Max. AC Input/Output Current (A)	50/47.9
Max. Continuous AC Passthrough (grid to load) (A)	80
Peak Power (off-grid) (W)	1.5 times of rated power, 10s
Power Factor Adjustment Range	0.8 leading to 0.8 lagging
Rated input/output voltage	220/380V, 230/400V 0.85Un-1.1Un 3L+N+PE
Rated Input/Output Grid Frequency/Range(Hz)	50/45-55, 60/55-65
Total Current Harmonic Distortion THDi	<3% (of nominal power)
DC Injection Current	<0.5% In
Efficiency	
Max. Efficiency	98.5%
Euro Efficiency	98.0%
MPPT Efficiency	>99%
Equipment Protection	
Integrated	DC Reverse Polarity Protection, AC Output Overcurrent Protection, Thermal Protection, AC Output Overvoltage Protection, AC Output Short Circuit Protection, DC Component Monitoring, Arc Fault Circuit Interrupter (AFCI) (optional), Anti-islanding Protection, DC Switch, Insulation Impedance Detection, Residual Current Detection
Surge Protection Level	TYPE II(DC), TYPE II(AC)
Interface	
LCD/LED Display	LCD+LED
Communication Interface	RS485, RS232, CAN
Monitor Mode	GPRS/WIFI/Bluetooth/4G/LAN(optional)
General Data	
Operating Temperature Range (°C)	-40 to +60°C, >45°C Derating
Permissible Ambient Humidity	0-100%
Permissible Altitude	3000m
Noise (dB)	≤55
Ingress Protection(IP) Rating	IP 65
Inverter Topology	Non-Isolated
Over Voltage Category	OVC II(DC), OVC III(AC)
Cabinet Size (WxHxD mm)	448×688×270 (Excluding Connectors and Brackets)
Weight (kg)	46
Type of Cooling	Intelligent Air Cooling
Warranty	5 Years / Up to 10 Years* *Warranty period varies by installation country. Refer to the Warranty Policy for details.
Grid Regulation	IEC 61727, IEC 62116, CEI 0-21, EN 50549, NRS 097, RD 140, UNE 217002, OVE-Richtlinie R25, G99, VDE-AR-N 4105
Safety / EMC Standard	IEC/EN 61000-6-1/2/3/4, IEC/EN 62109-1, IEC/EN 62109-2

Three Phase Hybrid Inverter

SUN-50K-SG01HP3-EU-BM4



Efficient

- High voltage battery, higher efficiency

Intelligent

- 6 time periods for battery charging/discharging

Flexible

- AC couple to retrofit existing solar system
- Max. 10 pcs parallel for on-grid and off-grid operation; support multiple batteries parallel

Reliable

- Max. charging/discharging current of 100A
- Support storing energy from diesel generator
- 100% unbalanced output

Technical Data

Model	SUN-50K-SG01 HP3-EU-BM4
Battery Input Data	
Battery Type	Lithium-ion
Battery Voltage Range (V)	160-800
Max. Charging/Discharging Current (A)	50+50
Charging Strategy for Li-ion Battery	Self-adaption to BMS
Number of Battery Input	2
PV String Input Data	
Max. PV access power (W)	100000
Max. PV Input Power (W)	80000
Max. PV Input Voltage (V)	1000
Start-up Voltage (V)	180
MPPT Voltage Range (V)	150-850
Rated PV Input Voltage (V)	600
Max. Operating PV Input Current (A)	36+36+36+36
Max. Input Short-Circuit Current (A)	55+55+55+55
No. of MPP Trackers/ No. of Strings MPP Tracker	4/2+2+2+2
AC Input/Output Data	
Rated AC Input/Output Active Power (W)	50000
Max. AC Input/Output Apparent Power (VA)	55000
Rated AC Input/Output Current (A)	75.8/72.5
Max. AC Input/Output Current (A)	83.4/79.8
Max. Continuous AC Passthrough (grid to load) (A)	200
Peak Power (off-grid) (W)	1.5 times of rated power, 10s
Power Factor Adjustment Range	0.8 leading to 0.8 lagging
Rated input/output voltage	220/380V,230/400V 0.85Un-1.1Un 3L+N+PE
Rated Input/Output Grid Frequency/Range(Hz)	50/45-55, 60/55-65
Total Current Harmonic Distortion THDi	<3% (of nominal power)
DC Injection Current	<0.5% In
Efficiency	
Max. Efficiency	97.6%
Euro Efficiency	97.0%
MPPT Efficiency	>99%
Equipment Protection	
Integrated	DC Reverse Polarity Protection, AC Output Overcurrent Protection, Thermal Protection, AC Output Overvoltage Protection, AC Output Short Circuit Protection, DC Component Monitoring, Arc Fault Circuit Interrupter (AFCI) (optional), Anti-islanding Protection, DC Switch, Insulation Impedance Detection, Residual Current Detection
Surge Protection Level	TYPE II(DC), TYPE II(AC)
Interface	
LCD/LED Display	LCD
Communication Interface	WIFI, RS485, CAN
Monitor Mode	GPRS/WIFI/Bluetooth/4G
General Data	
Operating Temperature Range (°C)	-40 to +60°C, >45°C Derating
Permissible Ambient Humidity	0-100%
Permissible Altitude	2000m
Noise (dB)	≤65
Ingress Protection(IP) Rating	IP 65
Inverter Topology	Non-Isolated
Over Voltage Category	OVC II(DC), OVC III(AC)
Cabinet Size (WxHxD mm)	527×894×294 (Excluding Connectors and Brackets)
Weight (kg)	80
Type of Cooling	Intelligent Air Cooling
Warranty	5 Years / Up to 10 Years* *Warranty period varies by installation country. Refer to the Warranty Policy for details.
Grid Regulation	IEC 61727, IEC 62116, CEI 0-21, EN 50549, NRS 097, RD 140, UNE 217002, OVE-Richtlinie R25, G99, VDE-AR-N 4105
Safety / EMC Standard	IEC/EN 61000-6-1/2/3/4, IEC/EN 62109-1, IEC/EN 62109-2

Deye

Three Phase Hybrid Inverter

SUN-50K-SG02HP3-EU-BM4-P



Efficient

- High voltage battery, higher efficiency

Intelligent

- 6 time periods for battery charging/discharging

Flexible

- AC couple to retrofit existing solar system
- Max. 10 pcs parallel for on-grid and off-grid operation; support multiple batteries parallel

Reliable

- Max. charging/discharging current of 160A
- Support storing energy from diesel generator
- 100% unbalanced output

Technical Data

Model	SUN-50K-SG02 HP3-EU-BM4-P
Battery Input Data	
Battery Type	Lithium-ion
Battery Voltage Range (V)	160-800
Max. Charging/Discharging Current (A)	160
Charging Strategy for Li-ion Battery	Self-adaption to BMS
Number of Battery Input	1
PV String Input Data	
Max. PV access power (W)	100000
Max. PV Input Power (W)	80000
Max. PV Input Voltage (V)	1000
Start-up Voltage (V)	180
MPPT Voltage Range (V)	150-850
Rated PV Input Voltage (V)	600
Max. Operating PV Input Current (A)	36+36+36+36
Max. Input Short-Circuit Current (A)	55+55+55+55
No. of MPP Trackers/ No. of Strings MPP Tracker	4/2+2+2+2
AC Input/Output Data	
Rated AC Input/Output Active Power (W)	50000
Max. AC Input/Output Apparent Power (VA)	55000
Rated AC Input/Output Current (A)	75.8/72.5
Max. AC Input/Output Current (A)	83.4/79.8
Max. Continuous AC Passthrough (grid to load) (A)	200
Peak Power (off-grid) (W)	1.5 times of rated power, 10s
Power Factor Adjustment Range	0.8 leading to 0.8 lagging
Rated input/output voltage	220/380V,230/400V 0.85Un-1.1Un 3L+N+PE
Rated Input/Output Grid Frequency/Range(Hz)	50/45-55, 60/55-65
Total Current Harmonic Distortion THDi	<3% (of nominal power)
DC Injection Current	<0.5% In
Efficiency	
Max. Efficiency	97.6%
Euro Efficiency	97.0%
MPPT Efficiency	>99%
Equipment Protection	
Integrated	DC Reverse Polarity Protection, AC Output Overcurrent Protection, Thermal Protection, AC Output Overvoltage Protection, AC Output Short Circuit Protection, DC Component Monitoring, Arc Fault Circuit Interrupter (AFCI) (optional), Anti-islanding Protection, DC Switch, Insulation Impedance Detection, Residual Current Detection
Surge Protection Level	TYPE II(DC), TYPE II(AC)
Interface	
LCD/LED Display	LCD
Communication Interface	WIFI/RS485/CAN
Monitor Mode	GPRS/WIFI/Bluetooth/4G
General Data	
Operating Temperature Range (°C)	-40 to +60°C, >45°C Derating
Permissible Ambient Humidity	0-100%
Permissible Altitude	3000m
Noise (dB)	≤65
Ingress Protection(IP) Rating	IP 65
Inverter Topology	Non-Isolated
Over Voltage Category	OVC II(DC), OVC III(AC)
Cabinet Size (WxHxD mm)	528×793×278 (Excluding Connectors and Brackets)
Weight (kg)	65
Type of Cooling	Intelligent Air Cooling
Warranty	5 Years / Up to 10 Years* *Warranty period varies by installation country. Refer to the Warranty Policy for details.
Grid Regulation	IEC 61727, IEC 62116, CEI 0-21, EN 50549, NRS 097, RD 140, UNE 217002, OVE-Richtlinie R25, G98, G99, VDE-AR-N 4105
Safety / EMC Standard	IEC/EN 61000-6-1/2/3/4, IEC/EN 62109-1, IEC/EN 62109-2

Three Phase Hybrid Inverter

SUN-80K-SG02HP3-EU-EM6



Efficient

- High voltage battery, higher efficiency

Intelligent

- 6 time periods for battery charging/discharging

Flexible

- AC couple to retrofit existing solar system
- Max. 10 pcs parallel for on-grid and off-grid operation; support multiple batteries parallel

Reliable

- Max. charging/discharging current of 160A
- Support storing energy from diesel generator

Technical Data

Model	SUN-80K-SG02HP3 -EU-EM6
Battery Input Data	
Battery Type	Lithium-ion
Battery Voltage Range (V)	160-1000
Max. Charging/Discharging Current (A)	80+80
Charging Strategy for Li-ion Battery	Self-adaption to BMS
Number of Battery Input	2
PV String Input Data	
Max. PV access power (W)	160000
Max. PV Input Power (W)	128000
Max. PV Input Voltage (V)	1000
Start-up Voltage (V)	180
MPPT Voltage Range (V)	150-850
Rated PV Input Voltage (V)	650
Max. Operating PV Input Current (A)	36+36+36+36+36+36
Max. Input Short-Circuit Current (A)	54+54+54+54+54+54
No. of MPP Trackers/ No. of Strings MPP Tracker	6/2+2+2+2+2+2
AC Input/Output Data	
Rated AC Input/Output Active Power (W)	80000
Max. AC Input/Output Apparent Power (VA)	88000
Rated AC Input/Output Current (A)	121.3/116
Max. AC Input/Output Current (A)	133.4/127.6
Max. Continuous AC Passthrough (grid to load) (A)	200
Peak Power (off-grid) (W)	1.5 times of rated power, 10s
Power Factor Adjustment Range	0.8 leading to 0.8 lagging
Rated input/output voltage	220/380V,230/400V 0.85Un-1.1Un 3L+N+PE
Rated Input/Output Grid Frequency/Range(Hz)	50/45-55, 60/55-65
Total Current Harmonic Distortion THDi	<3% (of nominal power)
DC Injection Current	<0.5% In
Efficiency	
Max. Efficiency	98.70%
Euro Efficiency	98.10%
MPPT Efficiency	>99%
Equipment Protection	
Integrated	DC Reverse Polarity Protection, AC Output Overcurrent Protection, Thermal Protection, AC Output Overvoltage Protection, AC Output Short Circuit Protection, DC Component Monitoring, Arc Fault Circuit Interrupter (AFCI) (optional), Anti-islanding Protection, DC Switch, Insulation Impedance Detection, Residual Current Detection
Surge Protection Level	TYPE II(DC), TYPE II(AC)
Interface	
LCD/LED Display	LCD+LED
Communication Interface	RS485,RS232,CAN
Monitor Mode	GPRS/WIFI/Bluetooth/4G/LAN(optional)
General Data	
Operating Temperature Range (°C)	-40 to +60°C, >45°C Derating
Permissible Ambient Humidity	0-100%
Permissible Altitude	3000m
Noise (dB)	≤65
Ingress Protection(IP) Rating	IP 65
Inverter Topology	Non-Isolated
Over Voltage Category	OVC II(DC), OVC III(AC)
Cabinet Size (WxHxD mm)	606×927×314 (Excluding Connectors and Brackets)
Weight (kg)	105
Type of Cooling	Smart Cooling
Warranty	5 Years / Up to 10 Years* *Warranty period varies by installation country. Refer to the Warranty Policy for details.
Grid Regulation	IEC 61727, IEC 62116, CEI 0-21, EN 50549, NRS 097, RD 140, UNE 217002, OVE-Richtlinie R25, G99, VDE-AR-N 4105
Safety / EMC Standard	IEC/EN 61000-6-1/2/3/4, IEC/EN 62109-1, IEC/EN 62109-2

Three Phase Hybrid Inverter

SUN-125K-SG02HP3-EU-GM10



Efficient

- High voltage battery, higher efficiency

Intelligent

- 6 time periods for battery charging/discharging

Flexible

- AC couple to retrofit existing solar system
- Max. 10 pcs parallel for on-grid and off-grid operation; support multiple batteries parallel

Reliable

- Max. charging/discharging current of 200A
- Support storing energy from diesel generator

Technical Data

Model	SUN-125K-SG02HP3-EU-GM10
Battery Input Data	
Battery Type	Lithium-ion
Battery Voltage Range (V)	160-1000
Max. Charging/Discharging Current (A)	100+100
Charging Strategy for Li-ion Battery	Self-adaption to BMS
Number of Battery Input	2
PV String Input Data	
Max. PV access power (W)	250000
Max. PV Input Power (W)	200000
Max. PV Input Voltage (V)	1000
Start-up Voltage (V)	180
MPPT Voltage Range (V)	150-850
Rated PV Input Voltage (V)	650
Max. Operating PV Input Current (A)	42+42+42+42+42+42+42+42+42+42
Max. Input Short-Circuit Current (A)	63+63+63+63+63+63+63+63+63+63
No. of MPP Trackers/ No. of Strings MPP Tracker	10/2+2+2+2+2+2+2+2+2+2
AC Input/Output Data	
Rated AC Input/Output Active Power (W)	125000
Max. AC Input/Output Apparent Power (VA)	135000
Rated AC Input/Output Current (A)	189.4/181.2
Max. AC Input/Output Current (A)	204.6/195.7
Max. Continuous AC Passthrough (grid to load) (A)	250
Peak Power (off-grid) (W)	1.5 times of rated power, 10s
Power Factor Adjustment Range	0.8 leading to 0.8 lagging
Rated input/output voltage	220/380V, 230/400V 0.85Un-1.1Un 3L+N+PE
Rated Input/Output Grid Frequency/Range(Hz)	50/45-55, 60/55-65
Total Current Harmonic Distortion THDi	<3% (of nominal power)
DC Injection Current	<0.5% In
Efficiency	
Max. Efficiency	98.70%
Euro Efficiency	98.10%
MPPT Efficiency	>99%
Equipment Protection	
Integrated	DC Reverse Polarity Protection, AC Output Overcurrent Protection, Thermal Protection, AC Output Overvoltage Protection, AC Output Short Circuit Protection, DC Component Monitoring, Arc Fault Circuit Interrupter (AFCI) (optional), Anti-islanding Protection, DC Switch, Insulation Impedance Detection, Residual Current Detection
Surge Protection Level	TYPE II(DC), TYPE II(AC)
Interface	
LCD/LED Display	LCD+LED
Communication Interface	RS485,RS232,CAN
Monitor Mode	GPRS/WIFI/Bluetooth/4G/LAN(optional)
General Data	
Operating Temperature Range (°C)	-40 to +60°C, >45°C Derating
Permissible Ambient Humidity	0-100%
Permissible Altitude	3000m
Noise (dB)	≤65
Ingress Protection(IP) Rating	IP 65
Inverter Topology	Non-Isolated
Over Voltage Category	OVC II(DC), OVC III(AC)
Cabinet Size (WxHxD mm)	734×1091×344 (Excluding Connectors and Brackets)
Weight (kg)	161.7
Type of Cooling	Intelligent Air Cooling
Warranty	5 Years / Up to 10 Years* *Warranty period varies by installation country. Refer to the Warranty Policy for details.
Grid Regulation	IEC 61727,IEC 62116,CEI 0-21,EN 50549, NRS 097,RD 140,UNE 217002, OVE-Richtlinie R25,G99,VDE-AR-N 4105
Safety / EMC Standard	IEC/EN 61000-6-1/2/3/4, IEC/EN 62109-1, IEC/EN 62109-2

125kW-2.5MW C&I ESS SOLUTION

PCS with Integrated EMS, Supporting Flexible Integration with MPPT & STS Modules



Power Continuity

- **UPS-Level Switching:**
<10 ms seamless on-grid/off-grid switching via STS.

- **Strong Overload Capability:**
Supports 2 × peak power output for 12s in off-grid mode, ensuring stable operation of critical loads.

Smart Energy Management

- **Integrated EMS:**
Supports zero-export, TOU optimization, peak shaving and demand management.

- **Flexible Control:**
Flexible Control: Local touchscreen control and remote monitoring via Deye Cloud.

System Scalability

- **Flexible Expansion:**
100/125kW PCS units enable system scalability up to 2.5 MW.

- **Large Energy Storage:**
Up to 257 kWh per battery cluster, supporting up to 16 clusters per PCS.

Smart Battery Management

- **Long Backup Duration:**
Supports up to 32 hours of continuous backup power.

- **Intelligent Balancing:**
Independent BMS optimizes charge/discharge distribution, extending overall battery lifespan.

Efficient PV Integration

- **Multi-Tracker MPPT Design:**
Each MPPT module supports 8 MPPT trackers, with up to 40 A per tracker.

- **High PV Capacity:**
Up to 200 kWp PV capacity per MPPT module.

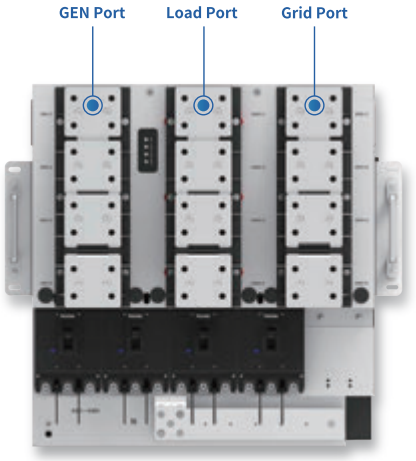
Reliability & Protection

- **IP65 Design:**
IP65-rated PCS and MPPT modules for reliable operation in demanding environments.

- **Advanced Battery Safety:**
$LiFePO_4$ cells with integrated aerosol fire suppression for enhanced protection.

120kW-2.5MW C&I ESS Solution

STS Module



- Three independent power paths (diesel / load / grid), with each path supporting 500 kW.
- <10 ms on-grid/off-grid switching*.
- 5×100 kW or 4×125 kW PCS units per STS module, up to 5 STS modules in parallel.



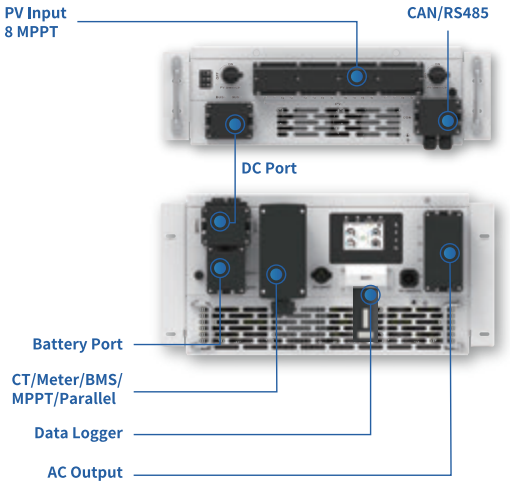
MPPT Module & PCS Module

MPPT Module (8 MPPT Trackers)

- 8 MPPT Trackers, up to 40A per tracker.
- Up to 200 kWp PV capacity per MPPT module.

PCS Module (125kW)

- **200A** charge and discharge current.
- Maximum efficiency of **98.5%**.
- System rated power up to **2.5MW**.
- Supports instantaneous peak power up to 200% of rated power.
- Integrates zero-export and time-of-use control functions.
- Built-in EMS for intelligent energy management and optimized system operation.



* For stable off-grid operation, the continuous off-grid load should not exceed 70% of the PCS rated power.

120kW-2.5MW C&I ESS Solution

MPPT Module	SUN-MPPT-L01-EU-AM8
PV String Input Data	
Max. PV access power(kW)	200
Max. PV Input Power(kW)	160
Max. PV Input Voltage (V)	800
Start-up Voltage (V)	200
MPPT Voltage Range (V)	180-750
Full Load MPPT Voltage Range (V)	450-750
Rated PV Input Voltage (V)	600
Max. Operating PV Input Current (A)	40+40+40+40+40+40+40+40
Max. Input Short-Circuit Current (A)	60+60+60+60+60+60+60+60
No. of MPP Trackers	8
Efficiency	
Max. Efficiency	>99%
MPPT Efficiency	>99.9%
Equipment Protection	
DC input reverse protection	YES
DC ARC protection	Optional
Anti-PID(Potential Induced Degradation)	Optional
DC Switch	YES
Surge Protection Level	TYPE II
General Data	
Ingress Protection(IP) Rating	IP65
Over Voltage Category	OVC II
Cabinet Size[W×H×D] (mm)	543x198x700
Weight (kg)	41.75
Type Of Cooling	Intelligent air cooling
Safety EMC/Standard	IEC/EN 62109-1
DC Output Data	
DC Output Voltage Range(V)	630-1000
Max. DC Output Current(A)	200

STS Module	SUN-STS500L
Grid/PCS Side Data	
Rated AC Input/Output Active Power (kW)	500
Rated AC Input/Output Current (A)	758/725
Rated Input/Output Voltage(V)	220/380, 230/400 (three phase)
Grid Connection Form	3L/N/PE
Rated Input/Output Grid Frequency	50Hz/60Hz
Load Side Data	
Rated Output Active Power (kW)	500
Rated Output Current (A)	758/725
Rated Output Voltage(V)	220/380, 230/400 (three phase)
Grid Connection Form	3L/N/PE
Rated Output Grid Frequency	50Hz/60Hz
GEN Side Data	
Rated AC Input Active Power (kW)	500
Rated AC Input Current (A)	758/725
Rated Input Voltage(V)	220/380, 230/400 (three phase)
Grid Connection Form	3L/N/PE
Rated Input Grid Frequency	50Hz/60Hz
General Data	
Off grid switching time	<10ms
Ingress Protection(IP) Rating	IP20
Over Voltage category	OVC III
Cabinet Size[W×H×D] (mm)	543x575x671
Weight (kg)	108
Type Of Cooling	Natural Cooling
Safety EMC/Standard	IEC/EN 61439-1/-2

PCS Model	SUN-125K-PCS01HP3
Battery Data	
Battery Type	Lithium-ion
Battery Voltage Range (V)	630-1000
Max. Charging Current (A)	200
Max. Discharging Current (A)	200
Charging Strategy for Li-ion Battery	Self-adaption to BMS
Number of Battery Input	1
DC Input Data	
DC Input Voltage Range(V)	630-1000
Max. DC Input Current(A)	200
AC Input/Output Data	
Rated AC Input/Output Active Power (kW)	125
Max. AC Input/Output Apparent Power (kVA)	125
Rated AC Input/Output Current (A)	189.4/181.2
Max. AC Input/Output Current (A)	189.4/181.2
Rated Input/Output Voltage/Range(V)	220/380, 230/400 0.85Un-1.1Un
Grid Connection Form	3L+N+PE
Rated Input/Output Grid Frequency/Range	50Hz/45Hz-55Hz 60Hz/55Hz-65Hz
Power Factor Adjustment Range	-1~1
Total Current Harmonic Distortion THDi	<3% (of nominal power)
DC Injection Current	<0.5% In
Efficiency	
Max. Efficiency	98.5%
Euro Efficiency	97.8%
Equipment Protection	
Integrated	AC Output Overcurrent Protection, AC Output Overvoltage Protection, AC Output Short Circuit Protection, Thermal Protection,Anti-islanding Protection, Insulation Impedance Detection, Residual Current Detection
Surge Protection Level	TYPE II(DC), TYPE II(AC)
Interface	
LCD/LED display	LCD
Communication Interface	WIFI, RS485, CAN, Meter
General Data	
Operating Temperature Range(°C)	-40°C-60°C,>45°C Derating
Permissible Ambient Humidity	0-95%
Permissible Altitude	4000m
Noise	<75dB
Ingress Protection(IP) Rating	IP 65(PCS Module)
Cabinet Size[W×H×D] (mm)	543x310x775 (Excluding connectors and brackets)
Weight (kg)	81.86
Inverter Topology	Non-Isolated
Over Voltage Category	OVC II(DC), OVC III(AC)
Type of Cooling	Intelligent Air Cooling
Warranty	5 Years/10 Years the Warranty Period Depends the Final Installation Site of Inverter, More Info Please Refer to Warranty Policy
Grid Regulation	IEC 61727, IEC 62116, CEI 0-21, EN 50549, NRS 097, RD 140, UNE 217002, OVE-Richtlinie R25, G99, VDE-AR-N 4105
Safety/EMC Standard	IEC/EN 62477-1



Comprehensive Protection

- Advanced BMS with active fuse

Optimized Energy Density

- Integrated PACK: reduced line loss, enhanced energy density

Easy Maintenance

- Auto-networking, Local monitoring mode for battery, remote monitoring mode for ESS

Superior Performance

- Support Max. 1C charge & 1.2C discharge (SE-F5), GaN MOSFETs: 50% loss reduction, high-temp resistance

Flexible Expansion

- Max. 32 units in parallel

Reliable Durability

- Operates reliably in -20℃ to 55 ℃, natural cooling

SE-F5-C&SE-F12-C&SE-F16-C

Technical Data

Main Parameters

		SE-F5-C	SE-F12-C	SE-F16-C
Battery Chemistry			$LiFePO_4$	
Capacity		100 Ah	235 Ah	314 Ah
Scalability[1]			Max. 32 pcs in parallel	
Nominal Voltage			51.2 V	
Operating Voltage			44.8 V ~ 57.6 V	
Nominal Energy		5 12 kWh	12.03 kWh	16 kWh
Charge Current[2]	Recommend	50 A	115 A	157 A
	Max. Continuous	100 A	230 A	160 A
	Peak	120 A (10 sec)	280 A (10 sec)	
Discharge Current[2]	Recommend	50 A	115 A	157 A
	Max. Continuous	120 A	230 A	
	Peak	150 A (10 sec)	280 A (10 sec)	

Other Parameters

Model	SE-F5-C	SE-F12-C	SE-F16-C
Recommend Depth of Discharge	80% DoD	90% DoD	
Dimension (W × D × H, mm) (Without hanging board)	370 × 548 × 140	400 × 559 × 233	400 × 708 × 233
Weight Approximate	41 kg	84 kg	109 kg
LED Indicator	LED (SOC, working, protecting) & Buzzer		
IP Rating of Enclosure	IP21		
Operating Temperature	Charge: 0~55°C / Discharge: -20~55°C		
Storage Temperature	0~35°C		
Relative Humidity	95% (non-condensing)		
Altitude	≤3000m		
Cycle Life	≥6000(25°C±2°C,70%EOL)		
Installation	Wall-Mounted, Floor-Mounted, Stack-Mounted		
Communication	CAN2.0, RS485, Bluetooth+APP		
Warranty Period[3]	5 years	5 Years / 10 Years [4]	5 Years / 10 Years [4]
Energy Throughput[3]	8 MWh	37 MWh	50 MWh
Certification	UN38.3, MSDS, CE, CB	UN38.3, CE, CB	

Product Comparison

Model	Nominal Energy	Charge / Discharge Rate	DoD	Warranty	Size
SE-F5-C	5.12kWh, 51.2V, 100Ah	1C/1.2C	80%	5years	370 x 548 x 140 mm
SE-F12-C	11.8kWh, 51.2V, 235Ah	1C/1C	90%	10years	400 x 583 x 233 mm
SE-F16-C	16kWh, 51.2V, 314Ah	0.5C/0.7C	90%	10years	400 x 708 x 233 mm

[1] Max. 64 pcs can parallel with CAN-Bridge.
[2] Operating current is affected by temperature and SOC.
[3] Conditions apply, refer to Deye Warranty Letter.
[4] The warranty period depends on the final installation site of the battery system. For more information, please refer to the warranty policy.



Comprehensive Protection

- Advanced BMS with active fuse

Optimized Energy Density

- Integrated PACK: reduced line loss, enhanced energy density

Easy Maintenance

- Auto-networking, Local monitoring mode for battery, remote monitoring mode for ESS

Superior Performance

- GAN MOSFETs: 50% loss reduction, high-temp resistance

Flexible Expansion

- Max. 32 units in parallel

Reliable Durability

- Operates reliably in -20℃ to 55℃, natural cooling

SE-F5 Lite & SE-F12 Lite

Technical Data

Main Parameters

		SE-F5 Lite	SE-F12 Lite
Battery Chemistry		$LiFePO_4$	
Capacity		100 Ah	235 Ah
Scalability[1]		Max. 32 pcs in parallel	
Nominal Voltage		51.2 V	
Operating Voltage		44.8 V ~ 57.6 V	
Nominal Energy		5.12 kWh	12.03 kWh
Charge Current[2]	Recommend	33 A	75 A
	Max. Continuous	50 A	120 A
	Peak	60 A (10sec)	180 A (10sec)
Discharge Current[2]	Recommend	33 A	75 A
	Max. Continuous	60 A	140 A
	Peak	72 A (10 sec)	210 A (10 sec)

Other Parameters

Model	SE-F5 Lite	SE-F12 Lite
Recommend Depth of Discharge	80% DoD	95% DoD
Dimension (W × D × H, mm) (Without hanging board)	360 x 545 x 142.5	400 x 594 x 233
Weight Approximate	41 kg	84 kg
LED Indicator	LED (SOC, working, protecting) & Buzzer	
IP Rating of Enclosure	IP21	
Operating Temperature	Charge:0~55°C/Discharge:-20~55°C	
Storage Temperature	0~35°C	
Relative Humidity	95% (non-condensing)	
Altitude	≤3000m	
Cycle Life	≥6000(25°C±2°C,70%EOL)	
Installation	Wall-Mounted, Floor-Mounted, Stack-Mounted	
Communication	CAN2.0, RS485, Bluetooth+APP	
Warranty Period [3]	5 years	5 Years / 10 Years [4]
Energy Throughput [3]	8 MWh	37 MWh
Certification	UN38.3, CE, CB	

Product Comparison

Model	Nominal Energy	Charge / Discharge Rate	DoD	Warranty	Size
SE-F5 Lite	5.12kWh, 51.2V, 100Ah	0.5C/0.6C	80%	5 years	360 x 545 x 142.5mm
SE-F12 Lite	12.03kWh, 51.2V, 235Ah	0.5C/0.6C	95%	10 years	400 x 594 x 233mm

[1] Max. 64 pcs can parallel with CAN-Bridge.
[2] Operating current is affected by temperature and SOC.
[3] Conditions apply, refer to Deye Warranty Letter.
[4] The warranty period depends on the final installation site of the battery system. For more information, please refer to the warranty policy.



Indoor C&I ESS Solution

Rack-Mounted Battery (HV)

ORION B

Intelligent Control

- Peak-valley mgmt, anti-backflow, overload protection
  Load tracking, demand control, backup power, phase separation

Scalable

- Support up to 20 units in parallel, maximum 2MW/4.3MWh

Multi-Fusion

- Integrated EMS, PCS, and BMS
- Support expansion of MPPT module
- Support off-grid backup

Reliable

- Operating temp : -20°C to 55°C
- Operate up to 3000m altitude
- 1.1x overload capacity
- Balancing solutions extend battery life
- Triple auxiliary power design for stable supply

Easy Maintenance

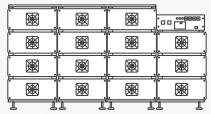
- 5U Standard Chassis
- User Interface & Bluetooth App
- USB & Cloud Upgrades
- TCP Protocol for EMS
- Fault Signal Input Support

Safer

- LFP batteries
- Support aerosol fire extinguishing

BOS-B Pro

Technical Data



Main Parameters

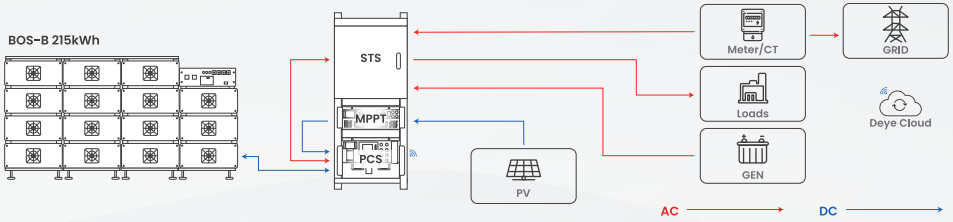
Battery Module Energy (kWh)		14.3
Battery Module Nominal Voltage (V)		51.2
Battery Module Capacity (Ah)		280
Module Weight Approximate (kg)		123
Battery Module Qty In Series (Optional)		15
Scalability	PCS or Inverter	14-15 units for PCS on-grid applications, 15 units for PCS off-grid applications, 5-15 units for hybrid inverter systems
	PCS + MPPT	15 units (on/off-grid) for MPPT Open-Circuit Voltage around 750V 14 units (on-grid) for MPPT Open-Circuit Voltage around 700V

Battery Model Number	BOS-B70	BOS-B85	BOS-B100	BOS-B110	BOS-B125	BOS-B140	BOS-B155	BOS-B170	BOS-B185	BOS-B200	BOS-B215
Battery Module Qty In Series (Optional)	5	6	7	8	9	10	11	12	13	14	15
System Nominal Voltage (V)	256	307.2	358.4	409.6	460.8	512	563.2	614.4	665.6	716.8	768
System Operating Voltage (V)	208~292	249.6~350.4	291.2~408.8	332.8~467.2	374.4~525.6	416~584	457.6~642.4	499.2~700.8	540.8~759.2	582.4~817.6	624~876
System Energy (kWh)	71.5	85.8	100.1	114.4	128.7	143	157.3	171.6	185.9	200.2	214.5
System Usable Energy (kWh)	64.35	77.22	90.09	102.96	115.83	128.7	141.57	154.44	167.31	180.18	193.05
Rated DC Power(kWH)	43.008	51.6096	60.2112	68.8128	77.4144	86.016	94.6176	103.2192	111.8208	120.4224	129.024
Max.Charge/Discharge Current (A)	168										

Other Parameters

Operating Temperature (°C)	discharge： -20 ~ 55 charge： 0 ~ 55										
Storage Temperature (°C)	0 ~ 35										
Thermal Management	Smart fan cooling										
LCD Display	SOC / Fault Code										
Status Indicator	Yellow : Battery High Voltage Power On Red : Battery System Alarm										
Communication Port	TCP / RS485 / CAN										
Communication With BMS	CAN										
Humidity	5% ~ 85%										
Altitude	⩽3000m										
IP Rating of Enclosure	IP20										
System Dimension (W × H × D, mm)	1067 × 1136 × 800			1608 × 1136 × 800				2150 × 1136 × 800			
System Weight Approximate (kg)	680	803	926	1066	1189	1312	1435	1575	1698	1821	1944
Installation Location	Rack Mounted										
Recommend Depth of Discharge	90%										
Cycle Life	25±2°C, 0.5C / 0.5C, EOL70%⩾6000										
Warranty Period	10 years										
Certification	CE / IEC62619 / IEC62040 / UN38.3										

Typical Application Scenario





Indoor C&I ESS Solution

Rack-Mounted Battery (HV)

ORION B

Intelligent Control

- Peak-valley mgmt, anti-backflow, overload protection
- Load tracking, demand control, backup power, phase separation

Scalable

- Support up to 16 units in parallel, maximum 2.3MW/4.1MWh

Multi-Fusion

- Integrated EMS, PCS, and BMS
- Support expansion of MPPT module
- Support off-grid backup

Reliable

- Operating temp : -20°C to 55°C
- Operate up to 3000m altitude
- 1.1x overload capacity
- Balancing solutions extend battery life
- Triple auxiliary power design for stable supply

Easy Maintenance

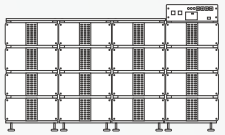
- 5U Standard Chassis
- User Interface & Bluetooth App
- USB & Cloud Upgrades
- TCP Protocol for EMS
- Fault Signal Input Support

Safer

- LFP batteries
- Support aerosol fire extinguishing

BOS-B Pro-A3

Technical Data



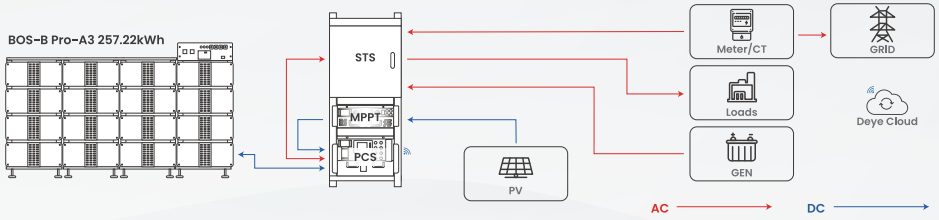
Main Parameters

Battery Module Energy (kWh)	16.08												
Battery Module Nominal Voltage (V)	51.2												
Battery Module Capacity (Ah)	314												
Module Weight Approximate (kg)	126												
Battery Module Qty In Series (Optional)	5-16												
Matching Mode	PCS or Hybrid Inverter	14-16 units for PCS on-grid applications, 15-16 units for PCS off-grid applications, 5-16 units for hybrid inverter systems											
	PCS + MPPT	16 units (on/off-grid) for MPPT Open-Circuit Voltage ≤800V 15 units (on/off-grid) for MPPT Open-Circuit Voltage ≤ 750V 14 units (on-grid) for MPPT Open-Circuit Voltage ≤ 700V											
Battery Model Number		BOS-B80 Pro-A3	BOS-B96 Pro-A3	BOS-B112 Pro-A3	BOS-B128 Pro-A3	BOS-B144 Pro-A3	BOS-B160 Pro-A3	BOS-B176 Pro-A3	BOS-B192 Pro-A3	BOS-B208 Pro-A3	BOS-B224 Pro-A3	BOS-B240 Pro-A3	BOS-B256 Pro-A3
Battery Module Qty In Series (Optional)		5	6	7	8	9	10	11	12	13	14	15	16
System Nominal Voltage (V)		256	307.2	358.4	409.6	460.8	512	563.2	614.4	665.6	716.8	768	819.2
System Operating Voltage (V)		208-292	249.6-350.4	291.2-408.8	332.8-467.2	374.4-525.6	416-584	457.6-642.4	499.2-700.8	540.8-759.2	582.4-817.6	624-876	665.6-934.4
System Energy (kWh)		80.3	96.4	112.5	128.6	144.6	160.7	176.8	192.9	208.9	225	241.1	257.2
System Usable Energy (kWh)		72.27	86.76	101.25	115.74	130.14	144.63	159.12	173.61	188.01	202.5	216.99	231.48
Max DC Power(kW)		46.08	55.296	64.512	73.728	82.944	92.16	101.376	110.592	119.808	129.024	138.24	147.456
Max.Charge/Discharge Current (A)		180											

Other Parameters

Operating Temperature (°C)	Charge : 0 ~ 55 Discharge : -20 ~ 55											
Storage Temperature (°C)	0 ~ 35											
Thermal Management	Smart fan cooling											
LCD Display	SOC / Fault Code											
Status Indicator	Yellow : Battery High Voltage Power On Red : Battery System Alarm											
Communication Port	TCP / RS485 / CAN											
Communication With BMS	CAN											
Humidity	5% ~ 85%											
Altitude	≤3000m											
IP Rating of Enclosure	IP20											
Noise (dB)	TBD											
System Dimension (W × H × D, mm)	1067 × 1305 × 800			1608 × 1305 × 800				2150 × 1305× 800				
System Weight Approximate (kg)	758	884	1010	1136	1310	1436	1562	1688	1862	1988	2114	2240
Installation Location	Rack Mounted											
Recommend Depth of Discharge	90%											
Cycle Life	25±2°C, 0.5C / 0.5C, EOL70%≥6000											
Warranty Period	10 years											
Certification	CE / IEC62619 / IEC62040 / UN38.3											

Typical Application Scenario





Small-Scale C&I ESS Solution

Rack-Mounted Battery (HV)

Safe and Reliable

- Intelligent BMS
- Firefighting module
- Dual electrode disconnection design
- Support up to 160A current output

Flexible Expansion

- Support 7 ~ 21 packs
- Inverter 50 ~ 100kW, Battery 54 ~ 161kWh
- Easy capacity expansion and save more budget

Smart Operation

- Cloud-based monitoring
- Keep track of the operating status
- Intelligent strategy control
- Effectively saving on electricity bill

Intelligent Control

- Protection against over-discharge, over-charge,
- over-current and extreme temperatures
- Automatically manage charge / discharge
- States and balances cell current voltage
- Uploading of battery data via TCP protocol

Easy Installation

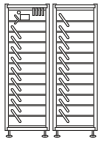
- 3U rack embedded design
- A concise data display interface
- Multiple battery modules can be in parallel for expanding
- Dual power output plugins, each supports 100A
- Connectable to two inverter DC interfaces
- USB, Bluetooth connection

Long Service Life

- 10-year warranty
- 6000 cycles

BOS-A

Technical Data



Main Parameters

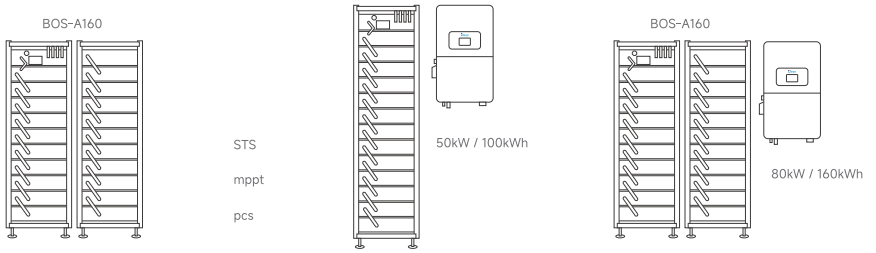
Cell Chemistry			$LiFePO_4$	
Module Energy (kWh)			7.68	
Module Nominal Voltage (V)			38.4	
Module Capacity (Ah)			200	
Module Dimension (W × D × H, mm)			601.5 × 520 × 135	
Module Weight Approximate (kg)			70	
Battery Module Qty In Series (Optional)		7	13	21
System Nominal Voltage (V)		268.8	499.2	806.4
System Operating Voltage (V)		235.2 ~ 306.6	436.8 ~ 569.4	705.6 ~ 919.8
System Energy (kWh)		53.76	99.84	161.28
System Usable Energy (kWh) [1]		48.38	89.85	145.15
Charge / Discharge	Recommend		100	
Current (A) [2]	Max.		160	

Other Parameters

Working Temperature (°C)	Charge : 0 ~ 55 / Discharge : -20 ~ 55		
Status Indicator	Yellow : Battery High Voltage Power On Red : Battery System Alarm		
Communication Port	CAN2.0		
Humidity	5% ~ 85%RH		
Altitude	≤3000m		
IP Rating of Enclosure	IP20		
Dimension (W × D × H, mm)	610 × 610 × 1900	610 × 610 × 2350	(610 × 610 × 1900) × 2
Weight Approximate (kg)	525	934	1501
Installation Location	Rack-Mounted		
Storage Temperature (°C)	0 ~ 35		
Recommend Depth of Discharge	90%		
Cycle Life	≥6000 (25±2°C, 0.5C / 0.5C, EOL70%)		
Warranty Period	10 years		
Certification	CE / IEC 62619 / IEC 62040 / UN38.3 / VDE-2510		

1. DC Usable Energy, test conditions : 90%DOD, 0.3C charge & discharge at 25°C. System usable energy may vary due to system configuration parameters.
2. The current is affected by temperature and SOC.
3. The warranty is due whichever reached first of warranty period or life cycle power.

Backup Power Duration Plan	1 hours	2 hours		4 hours	
Hybrid inverter power	100kW	50kW	80kW	50kW	80kW
Battery model	BOS-A160	BOS-A100	BOS-A160	BOS-A100	BOS-A160
Number of batteries	1 pc	1 pc	1 pc	2 pcs	2 pcs





Small-Scale C&I ESS Solution

Rack-Mounted Battery (HV)

ORION G

Convenient

- Quick installation standard of 19-inch embedded designed module is comfortable for installation and maintenance.

Intelligent BMS

- It has protection functions including over-discharge, over-charge, over-current and over-high or low temperature. The system can automatically manage charge and discharge state and balance current and voltage of each cell.

Eco-friendly

- The whole module is non-toxic, non-polluting and environmentally friendly.

Safe and Reliable

- Cathode material is made from LiFePO4 with safety performance and long cycle life, The module has less self-discharge, up to 6 months without charging it on shelf, no memory effect, excellent performance of shallow charge and discharge.

Flexible Configuration

- Multiple battery modules can be in parallel for expanding capacity and power. Support USB upgrade, remote up grade (Compatible with Deye inverter).

Wide Temperature

- Working temperature range is from -20°C to 55°C, with excellent discharge performance and cycle life.

BOS-G Pro (EU, SA, AU)

Technical Data

Main Parameters

Cell Chemistry	$LiFePO_4$			
Module Energy (kWh)	5.12			
Module Nominal Voltage (V)	51.2			
Module Capacity (Ah)	100			
Battery Module Number	BOS-G25 Pro	BOS-G40 Pro	BOS-G60 Pro	BOS-G80-Pro
Battery Module Qty In Series (Optional)	5 (Min)	8	12	16 (Max)
System Nominal Voltage (V)	256	409.6	614.4	819.2
System Operating Voltage (V)	220~292	352~467.2	528~700.8	704~934.4
System Energy (kWh)	25.6	40.96	61.44	81.92
System Usable Energy (kWh)	23.04	36.86	55.3	73.72
Rated DC Power	25.6	40.96	61.44	81.92
Charge / Discharge Current (A)	Recommend	50		
	Max.	100		
	Peak Discharge (2 mins, 25°C)	125		

Other Parameters

Operating Temperature (°C)	Charge : 0 ~ 55 / Discharge : -20 ~ 55			
Status Indicator	Yellow : Battery High Voltage Power On Red : Battery System Alarm			
Communication Port	CAN2.0 / RS485			
Humidity	5% ~ 85%RH			
Altitude	≤3000m			
IP Rating of Enclosure	IP20			
System Dimension (W x D x H,mm)	530 × 602 × 1629		530 × 602 × 2219	1060 × 602 × 1629
System Weight Approximate (kg)	290	428	622	883
Installation Location	Rack Mounting			
Storage Temperature(°C)	0 ~ 35			
Recommend Depth of Discharge	90%			
Cycle Life	25±2°C, 0.5C / 0.5C, EOL70%≥6000			
Warranty Period	10 years			
Certification	UN38.3 / CE / CE-EMC / IEC62040 / CEC / VDE			

Inverter	1h	2h	3h	4h
SUN-30K-SG01HP3-EU-BM3	X6	X12	X9 X2	X12 X2
SUN-40K-SG01HP3-EU-BM4	X8	X8 X2	X12 X2	X11 X3
SUN-50K-SG01HP3-EU-BM4	X10	X10 X2	X10 X3	X10 X4
SUN-60K-SG02HP3-EU-EM4	X12	X12 X2	X12 X3	X16 X3
SUN-80K-SG02HP3-EU-EM6	X16	X16 X2	X16 X3	X16 X4



WS-L4300-BC-3-A (AS, AF, LATAM, EU)
WS-PCS1125-2-A & WS-TS1000-2-A (AS, AF, LATAM, EU)

Ultimate Safety

- 3+3 Fire Protection System
- 3+3 Electrical Safety Safeguards
- AC Leakage & DC Insulation detection
- High-Voltage Interlock

Seamless Scalability

- Modular Architecture
- Flexible 2/4h Energy Storage Solutions
- Compact Design, High Energy Density

Versatile Applications

- Peak Shaving & Energy Arbitrage
- Virtual Power Plant (VPP) Ready
- Off-Grid & Microgrid Capable
- PV and BESS DC Coupling
- Hybrid Solar-BESS-Diesel Systems

Smart Cloud Management

- AI-Powered Load Optimization
- 24/7 Remote Monitoring & O&M
- Real-Time Battery Health & Safety Alerts
- Cloud-Connected Ecosystem

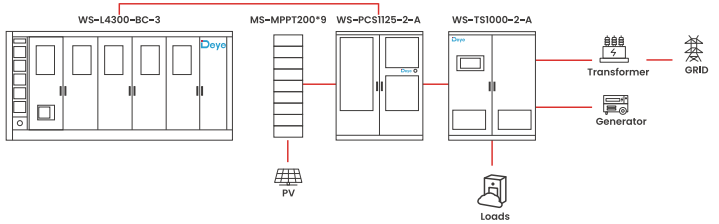
High Energy Density

- 4300kWh in 20' BESS container
- 1125kW PCS, 1800kW PV in Mppt Stacked Module

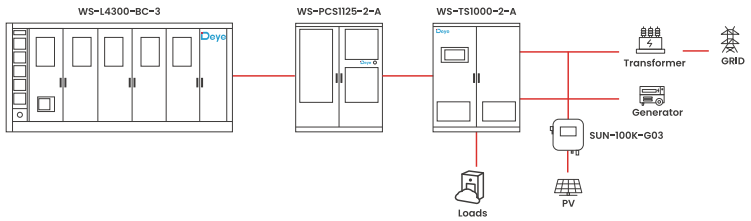
Cluster Management

- One Cluster Management, High Availability, More Friendly to the Cell

PV-BESS DC-Coupled Solution



PV-BESS AC-Coupled Solution



WS-L4300-BC-3-A (AS, AF, LATAM, EU)
WS-PCS1125-2-A & WS-TS1000-2-A (AS, AF, LATAM, EU)

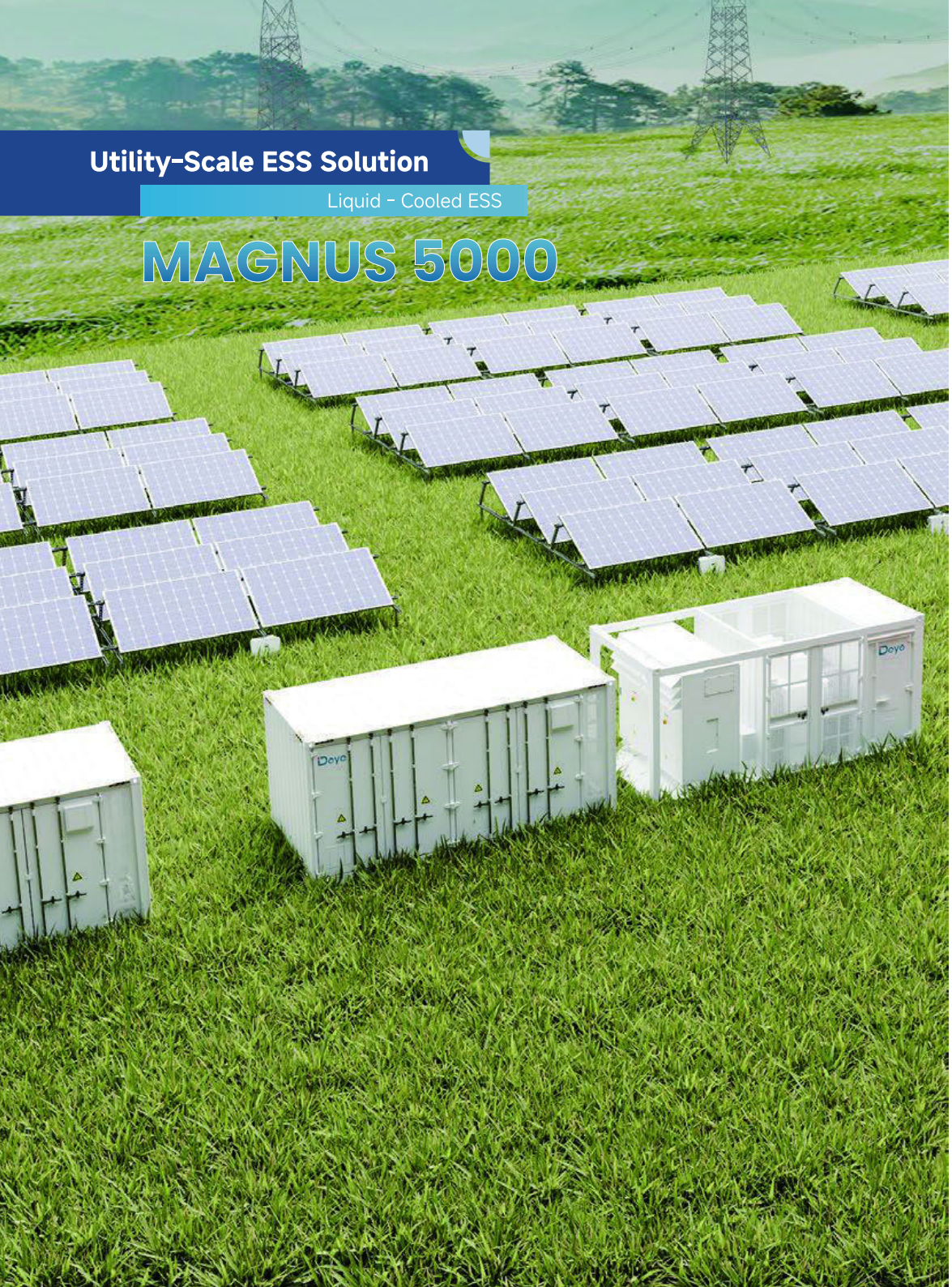
Technical Data

Liquid-Cooled ESS	
Model	**WS-L4300-BC-3-A**
DC Battery	
Battery Type	$LiFePO_4$
Nominal Capacity (Cell)	314Ah
Nominal Energy	4340kWh
PACK Configuration	1P48S
RACK Configuration	18 × 1P240S
Nominal DC Voltage	768Vdc
DC Voltage Range	648Vdc ~ 876Vdc
Charge and discharge rate	≤0.5P
Max. Charging / Discharging Current	3150A (18 × 175A)
No. of DC Output	18
System	
Operating Temperature	-30°C ~ +55°C
Storage Temperature	-30°C ~ +60°C
Humidity	0 ~ 95% (No condensation)
Type of cooling	Liquid cooling
Fire Suppression	Aerosol、Water
Ingress Protection	IP55
Anticorrosion grade	C4-M (Optional C5)
Altitude	≤2000m
Communication	CAN、RS485、TCP、DIDO
Weight	38000kg
Dimensions (W × D × H)	6058 × 2438 × 2896mm



Model	WS-PCS1125-2-A
PCS Data	
AC Rated Power	1125kW (9 × 125kW)
AC Rated Voltage / Frequency	400Vac / 50Hz (3L+N+PE)
AC Rated Current	1624A (9 × 180.4A)
Power Factor	-1 ~ +1
Battery Input Voltage Range	630Vdc ~ 1000Vdc
Max. Charging / Discharging Current	1710A (9 × 190A)
General Data	
Operating Temperature	-30°C~ +55°C (Discharge derating>45°C; Charge derating >40°C)
Humidity	0~95% (No condensation)
Ingress Protection	IP54
Anticorrosion grade	C4-M (Optional C5)
Altitude	≤3000m
Weight	≤2000kg
Dimensions (W × D × H)	1850 X 1120 X 2450mm

Model	WS-TS1000-2-A
Grid Side Data	
Rated AC Input/Output Active Power (kw)	1000
Rated AC Input/Output Current (A)	1450
Rated Input/Output Voltage (V)	220/380, 230/400 (three phase)
Grid Connection Form	3L/N/PE
Rated input/Output Grid Frequency	50Hz/60Hz
Load Side Data	
Rated AC Input/Output Active Power (kw)	1000
Rated AC Input/Output Current (A)	1450
Rated Input/Output Voltage (V)	220/380, 230/400 (three phase)
Grid Connection Form	3L/N/PE
Rated input/Output Grid Frequency	50Hz/60Hz
GEN Side Data	
Rated AC Input/Output Active Power (kw)	1000
Rated AC Input/Output Current (A)	1450
Rated Input/Output Voltage (V)	220/380, 230/400 (three phase)
Grid Connection Form	3L/N/PE
Rated input/Output Grid Frequency	50Hz/60Hz
System Data	
Dimension (W x D x H,mm)	2000 × 1200 × 2450mm
Weight Appr. (kg)	1600
System Operating temperature range	-20°C~55°C
Max. working altitude (m)	≤3000
IP Rating of Enclosure	IP54
Switching Time (ms)	≤20



WS-L5000-BC-2 (AS, AF, LATAM, EU)

The Low LCOS Multi-intelligent Energy Storage Solution

Economical and Efficient

- 20-foot container 5016kWh
- Maximum efficiency 94.5%
- Shoulder-to-shoulder/back-to-back installation

Flexible and Convenient

- Liquid cooling circuit equipped with bidirectional shut-off valve
- Equipped with waste liquid collection device
- Single/dual circuit output, 2/4h application

Smart Cloud Management

- AI-Power Load Optimization
- 24/7 Remote Monitoring & O&M
- Real-Time Battery Health & Safety Alerts
- Cloud-Connected Ecosystem

Ultimate Safety

- 2+3 Level Fire Protection System
- 3+2+1 Level Electrical Safety Protection

WS-MV5000-2 (AS, AF, LATAM, EU)

Integrated Boost and Conversion Solution

High-efficiency Conversion

- Three-level topology with a maximum efficiency of 99%
- Intelligent air-water cooling with a maximum operating temperature of 60℃

Safe and Reliable

- IP54/C5 high protection class
- Online insulation monitoring for safety assurance
- Interlocks with BMS and EMS to support multiple system protection functions

Grid Support

- Fast active/reactive power response, supporting source-grid-load-storage dispatching
- Weak grid support, enabling stable operation with SCR=1.1
- Equipped with grid-forming functions such as inertia support and microsecond-level voltage building

Flexible Application

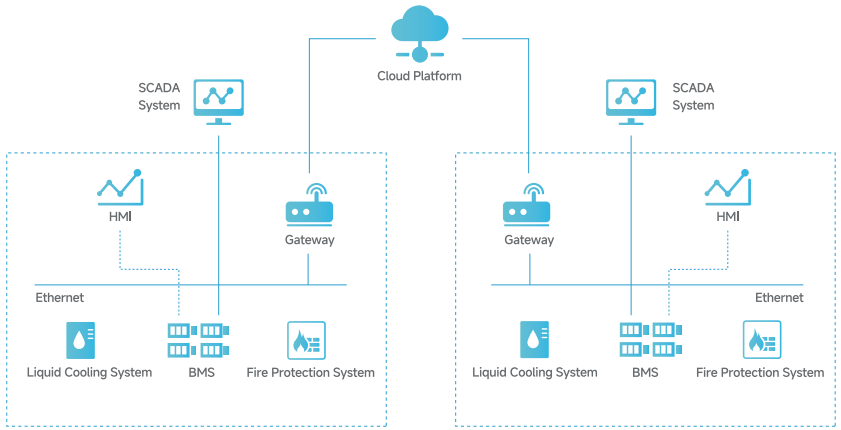
- Supports off-grid parallel operation and independently participates in grid black start
- Equipped with VSG, VF, PQ and other operating modes
- Supports application in various scenarios including high altitude, high salt spray, high temperature and extreme low temperature

WS-L5000-BC-2 (AS, AF, LATAM, EU)
WS-MV5000-2 (AS, AF, LATAM, EU)

Technical Data

Model	WS-L5000-BC-2
Battery Data	
Battery Type	$LiFePO_4$
Nominal Capacity (Cell)	314Ah
PACK Configuration	1P104S
RACK Configuration	4*1P104S
System Configuration	2*6P416S (12P416S)
Nominal Voltage	1331.2Vdc（1144-1497.6Vdc）
Nominal Energy	5016kWh（12× 418kWh）
Nominal Current	2*939A
Charge/Discharge Rate	0.5P
DC Efficiency	≥94.5%
System Data	
Operating Temperature	-30°C ~ +55°C（>45°C derating）
Storage Temperature	-30°C ~ +60°C
Humidity	0 ~ 95%RH
Type of Cooling BAT	Liquid cooling
Fire Suppression	Aerosol、Water
Ingress Protection	IP55
Anticorrosion Grade	C4/C5
Altitude	≤3000m
Weight	≤40500kg
Dimensions (W × D × H)	6058*2438*2896mm
Communication	Ethernet/CAN
Standard Compliance	IEC62619,IEC62477,IEC62933,IEC63056,IEC61000,UN38.3,UN3536,NFPA855,UL9540A

Architecture Diagram



Model	WS-MV5000-2
DC Side Data	
Max. DC Voltage	1500Vdc
DC Voltage Range for Nominal Power	1000Vdc~1500Vdc
Max. DC Current	1913A*4
No. of DC Inputs	4
PCS AC Side Data	
Rated output power	1250*4 kVA
Max output power	1250*3*4 kVA @35°C
Rated AC voltage	690Vac
AC voltage range	586.5~759Vac（-15%~+10%）
Max AC current	3138*4 A
Nominal grid frequency	50Hz
Grid frequency rage	45~55Hz
Power factor range	1 leading ~ 1 lagging
Efficiency	
Average Efficiency	≥98.3%
Transformer	
Isolation Mode	Oil-Immersed Transformer
Transformer Rated Power	5250kVA
LV/MV Voltage	0.69kV/20~35kV
Transformer Vector	Dy11
Cooling Type	ONAN
General Data	
Dimensions（W*D*H）	12192×2438×2896mm
Weight	≤40T
Protection Degree	IP65 for PCS，IP54 others
Cooling Method	Natural air cooling (transformer)/Air-water cooling (PCS)
Altitude	3000m(> 2000m derating)
Operating Temperature Range	-40°C~60°C (>45°C derating)
Humidity Range	0~100%（non-condensing）
Anticorrosion Grade	C4/C5
Over Voltage Range	DC OVC II / AC OVC III
Surge Protection	DC Type II / AC Type II
Communication	RS485/ETH/CAN
Standard Compliance	IEC 62477, IEC 61000, IEC 62920, IEC 61727, IEC 62116,IEC 60076, IEC 62271-100/103/200, EN 50549-2/-10,ANSI C57.12.10/20

* The warranty will be based on the warranty period or the minimum throughput of each model, whichever comes first.

C&I PV-BESS-EV CHARGING INTEGRATED SOLUTION

DEYE WINTER MC-LC SERIES



Intelligent Cloud Platform

- Customizable load algorithmic modules
- 24-hour online O&M
- Battery life and safety warning
- Device cloud interconnection

Versatile Expansion

- PCS/BMS/EMS All-in-one modular design
- Support up to 10 cabinets in parallel
- Support 2/4/6/8-hour energy storage applications
- Higher energy density to reduce footprint
- PV and BESS DC Coupling

Ultimate Safety

- 3+3 Fire Protection System
- 3+3 Electrical Safety Safeguards
- AC Leakage & DC Insulation Detection
- High-voltage interlocking, preventing loaded arc operation

Multiple Application Scenarios

- Peak-to-Valley arbitrage/Peak-to-Valley shifting
- Virtual power plant ready
- Off-grid operation (Islands, communication base stations, etc.)

5 Level
Extreme safety protection
Detection, early warning, smoke exhaust, fire fighting, explosion venting

10ms
Seamless on-grid and off-grid

480kW
DC fast charging
DC coupling for ESS and charging

C&I PV-BESS-EV Charging Integrated Solution

Model	MC-LC430-2H2	MC-LC430-BC-2 (DC BESS)
System parameter		
Operating Temperature	-25°C ~ +55°C	-25°C ~ +55°C
Storage Temperature	-30°C ~ +60°C	-30°C ~ +60°C
Humidity	0 ~ 95%	0 ~ 95%
Type of cooling	Liquid cooling	Liquid cooling
Fire Suppression	Aerosol, Water	Aerosol, Water
Ingress Protection	Overall IP54 (Battery compartment, Chiller control box: IP55; PCS: IP65)	Overall IP54 (Battery compartment, Chiller control box: IP55)
Anticorrosion grade	C4-M (Optional C5)	C4-M (C5M is optional)
Altitude	⩽2000m	⩽2000m
Communication	RS485, Modbus TCP, DIDO	RS485, Modbus TCP, DIDO
Weight	⩽5000kg	⩽4800kg
Dimensions (W × D × H)	2000 × 1350 × 2480mm (without air outlet cover)	2000 × 1350 × 2480mm
DC Data		
Battery	$LiFePO_4$	$LiFePO_4$
Nominal Capacity	280Ah	280Ah
Nominal Energy	430.08kWh	430.08kWh
Nominal DC Voltage	768Vd.c.	768Vd.c.
DC Voltage Range	648Vd.c ~ 876Vd.c.	648Vd.c ~ 876Vd.c.
Charge and discharge rate	charge 0.5P, discharge 1P	charge 0.5P, discharge 1P
AC Data		
Nominal AC Voltage	380/400V 3P+N+PE	
Rated Frequency	50 / 60Hz	
Rated Power	200kW	
Maximun Power	220kW (1.1 times of rated power)	
Power Factor	-1~1	

Model	MS-DC480-2 (480kW DC charge power cabinet)
DC Input Data	
Input standard	DC+ / DC- / PE
DC input voltage range	200Vdc ~ 850Vdc
DC input current range	⩽667A
DC rated input power	480kW @400Vdc⩽Vin⩽850Vdc
DC Output Data	
Dc output voltage range	150Vdc ~ 1000Vdc
Dc output current range	8 branches, max 250A each
Environmental Conditions	
Operating Temperature Range (°C)	-30°C to +50°C （derating above 50°C）
Storage Temperature (°C)	-40°C to +60°C
Humidity	⩽ 95%RH, no condensation
Cooling	Forced air cooling
Altitude	⩽ 2000 m (No derating required foroperation >1000m:Set the actual altitude value; the working temperature decreases-by 1°C for-every 100 m increase-in altitude).
IP Rating of Enclosure	IP54
Other Parameter	
Efficiency	⩾ 97.5%, @ peak full load
Dimension (W × H × D, mm)	1250mm× 2450mm (without eyebolt)× 1000mm
Approximate Weight (kg)	1200kg



Model	MS-DCC180-2 (DC charging terminal)
DC Input Data	
DC Input Voltage Range (V)	150~1000Vd.c. 500A max
Input number	2 DC inputs
DC Output Data	
Output interface	1 pile 2 guns, each gun 180kW charging power, support the CCS2 charging standard interface
DC Output range	150~1000Vd.c. 500A-max
Max. Output Power (W)	360kW(2*180kW).
Max. Output Current	Single gun Max.250A
Environmental Conditions	
Operating Temperature Range (°C)	-30°C to +55°C（derating above 50°C）
Storage Temperature (°C)	-40°C to +60°C
Humidity	⩽ 95%RH, no condensation
Cooling	Natural cooling
Altitude	⩽2000m
IP Rating of Enclosure	IP54
Other Parameter	
Dimension (W × H × D, mm)	1100 mm x 2200 mm (without GPS & 4G antenna) x 400mm
Approximate Weight (kg)	280kg

C&I PV-BESS-EV Charging Integrated Solution

Model	MS-MPPT400-2
System parameters	
Dimension (W × D × H, mm)	1000 × 1000 × 2480
Weight Appr. (kg)	⩽950kg
System Operating temperature range	-30°C ~ 45°C
Max. working altitude (m)	⩽2000m
IP Rating of Enclosure	IP54
STS parameters	
Rated insulation voltage (V)	DC1000
Rated working voltage (V)	AC400
Auxiliary equipment operating voltage (V)	AC220, DC24
Frequency	50/60Hz
Rated power of load (kW)	250
Rated power of the power grid (kW)	500
Rated power of oil engine (kW)	500
Switching Time	⩽10ms
MPPT parameters	
No. of MPPT	2
MAX.PV Access Power(kW)	400(200*2)
Max. PV Input Power (kW)	320(2*160)
Max. PV Input Voltage (V)	800
Start-up Voltage (V)	200
MPPT Voltage Range (V)	180 ~ 750
Full Load MPPT Voltage Range (V)	450 ~ 750
Rated PV Input Voltage (V)	600
Max. Operating PV Input Current (A)	2*(40+40+40+40+40+40+40+40)
Max. Input Short-Circuit Current (A)	2*(60+60+60+60+60+60+60+60)
No.of MPP Trackers	16（2*8）
Max. Efficiency	>99%
MPPT Efficiency	>99.9%

Model	MS-MPPT200-2
PV String Input Data	
MAX.PV Access Power(kW)	200
Max. PV Input Power (kW)	160
Max. PV Input Voltage (V)	800
Start-up Voltage (V)	200
MPPT Voltage Range (V)	180-750
Full Load MPPT Voltage Range (V)	450-750
Rated PV Input Voltage (V)	600
Max. Operating PV Input Current (A)	40+40+40+40+40+40+40+40
Max. Input Short-Circuit Current (A)	60+60+60+60+60+60+60+60
No. of MPP Trackers	8
Efficiency	
Max. Efficiency	>99%
MPPT Efficiency	>99.9%
Equipment Protection	
DC input reverse protection	YES
DC ARC protection	Optional
Anti-PID(Potential Induced Degradation)	Optional
DC Switch	YES
Surge Protection Level	TYPE II
General Data	
Ingress Protection(IP) Rating	IP65
Over Voltage Category	OVC I
Cabinet Size[W×H×D] (mm)	543x198x700
Weight (kg)	70
Type Of Cooling	Intelligent air cooling
Safety EMC/Standard	IEC/EN 62109-1
DC Output Data	
DC Output Voltage Range(V)	630-1000
Max. DC Output Current(A)	200



C&I Air-Cooled ESS | MS-EMS



AI Intelligence

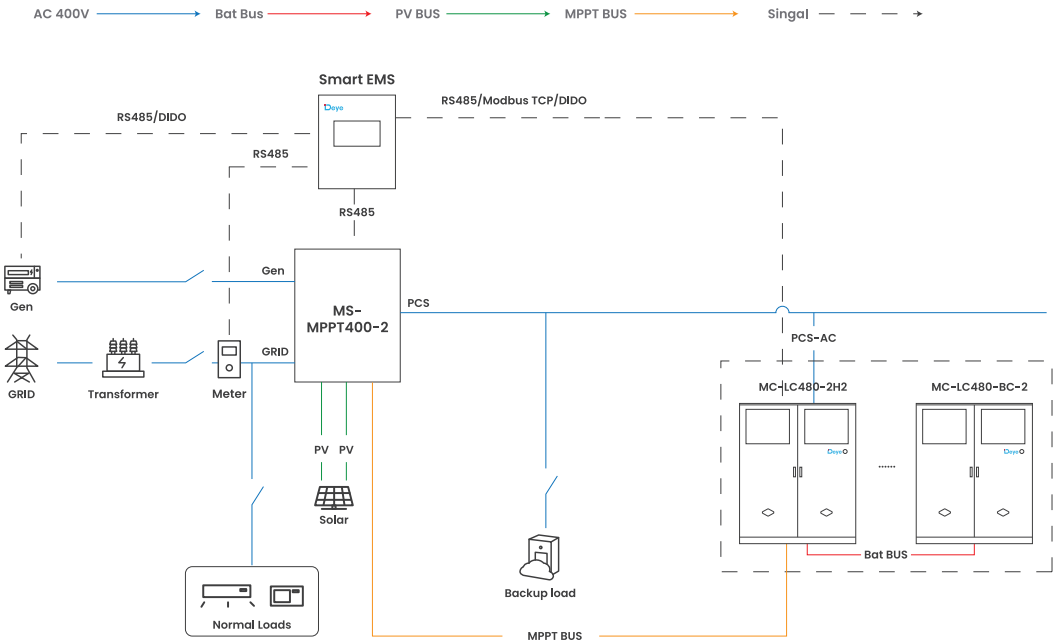
- Large capacity energy aggregation
- Real-time electricity price revenue calculation
- One click generation of statistical charts
- Maximum profit charging and discharging strategy

Efficient Operation and Maintenance

- Provide local / cloud operation and maintenance to ensure stable device operation
- Combination of multiple operation and maintenance methods for WEB / APP

Safe and Reliable

- Real-time alarm for equipment malfunction
- Support SOC balance management



C&I Air-Cooled EMS

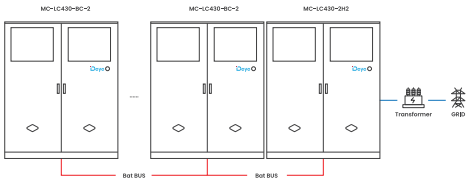
Model	MS-EMS
System	
Configuration	EMS controller, lightning protection module, switch power supply, UPS module, switch
Function	Basic functions : peak valley arbitrage, anti backflow, main transformer overload protection, load tracking, demand control, backup power function, phase separation control, SOC balancing, Deye Cloud monitoring Advanced features : load forecasting, production planning, electricity price planning, optimal economic curv
Communicate	
Ethernet (5 channels)	10 / 100 / 1000 Mbps
Fiber optic port (2 channels)	1Gbps
USB (2 channels)	Host
CAN (3 channels)	Isolation, with 2 channels supporting CAN-FD
RS485 (8 channels)	Isolation
RS232 (3 channels)	2 isolated channels, 1 non isolated debugging channel (DB9 socket)
TF Card (1 channel)	Standard TF card holder
LVDS (1 channel)	The physical interface is DVI (including 1 USB for touch)
M. 2 Interfaces (1 channel)	PCIe2.0 X1, Scalable SSD (standard 1TB)
MiniPCIe Interface (1 channel)	4G card with expandable USB communication protocol (standard)
Nano SIM Interface (1 channel)	Used in conjunction with the miniPCIe expansion 4G module
DI (17 channels)	Optocoupler isolation
DO (8 channels)	Relay isolation
WLAN	802.11 b / AC g n，HT 20 / 40，2.4 GHz 5 Ghz
4G Antenna	Support multi country frequency bands
Power Supply	
Communication Input	220Vac
DC IN	24Vdc
UPS Backup Power	24Vdc
Consumption	Max 25W
Environmental parameters	
Operation Temperature	-15°C ~ +50°C
Storage Temperature	-15°C ~ +50°C
Working Humidity	5% ~ 95%
Max. Working Altitude (m)	≤3000m
IP Rating of Enclosure	IP54
Anti-Corrosion Grade	C4-M (Optional C5)
Mechanical parameters	
Dimension (W × D × H, mm)	488 × 188 × 588
Weight Appr. (kg)	≤24.5kg
Installation Location	Indoor or outdoor, wall-mounted
Box Material	Metal
Incoming Specifications	AC power cord : Recommended wire diameter 1.5mm^2 DC power cord : Recommended wire diameter of 1.5mm^2 Eight core Ethernet cable: Recommended CAT5e Ethernet cable RS485 : Recommended 0.75mm^2~1.5mm^2 outdoor UV protection with shielding layer twisted pair cable length<1000m (baud rate 9600)

Versatile C&I Energy Storage Solutions

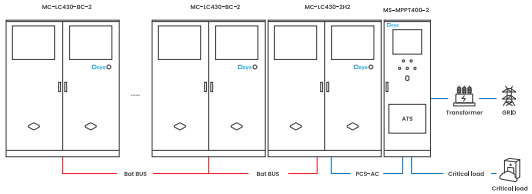
NOTE：MAX 3 battery cabinets（without PCS） parallel

AC 400V — Bat Bus — PV BUS — MPPT BUS

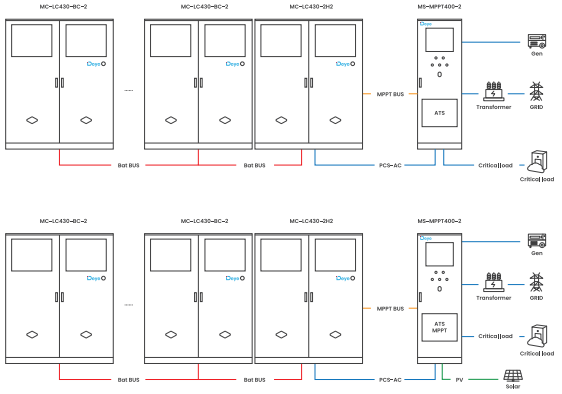
For ESS on-grid application



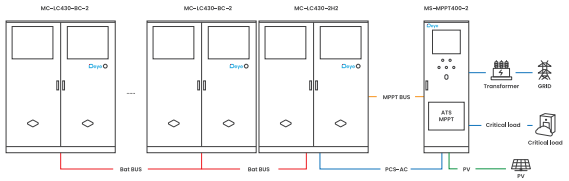
For backup power application



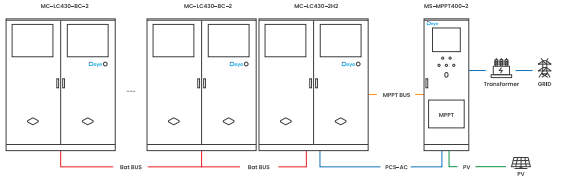
For backup power application with generator and grid



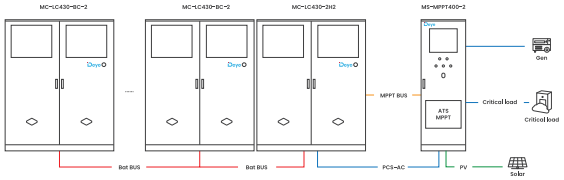
For backup power application with solar



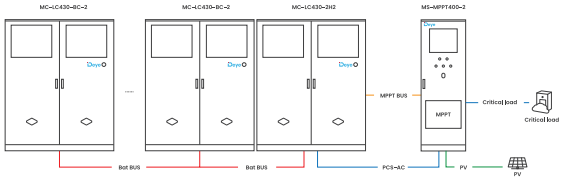
For ESS on-grid application with solar



For ESS off-grid application with solar and generator



For ESS off-grid application with solar



Integrated energy storage and charging application

Support up to four sets of double-gun charging terminals
Split type DC fast charging, With a maximum DC charging power of up to 180kW for a single gun
Supports flexible charging power distribution Adaptable to CCS2 charging interfaces
To solve the problem of insufficient capacity of new energy vehicles to access the distribution grid

